Relational Capacity Dynamics, Relational Foliation Gravity, and RFG-R

Full proof version: original branch, regular effective completion, local-GR matching, an extended-EFT likelihood boundary, an exact KGB posterior, a local ΛCDM reference posterior, and production Bayesian evidence

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Abstract

We present a unified theoretical framework that begins with a non-geometric macroscopic scalar theory—Relational Capacity Dynamics (RCD)—and constructs a covariant preferred-foliation completion—Relational Foliation Gravity (RFG). The objective is not only to compare a modified expansion history with ΛCDM, but to test whether a phenomenological dark-sector model can be replaced by a deeper dynamical account with explicit variables, actions, constraints, local limits, observables, likelihoods, and falsification gates. The primitive variable of the scalar sector is a strictly positive relational-capacity field $\mathcal{C}(\tau, \mathbf{x})$. We supply the complete ontology, dimensional structure, conservative action, full variational derivation, Hamiltonian formulation, Noether identities with explicit local currents, finite-response transport, emergent physical-time map, and linear stability analysis. The scalar sector is ghost-free for $\rho_{\mathcal{C}} > 0$, gradient-stable for $\kappa_{\mathcal{C}} > 0$, and strictly hyperbolic with characteristic speed $c_{\mathcal{C}} = \sqrt{\kappa_{\mathcal{C}}/\rho_{\mathcal{C}}}$.

The original covariant completion is defined by a metric–khronon action whose two free functions are uniquely reconstructed from a conserved relational-capacity law and the requirement of luminal tensor propagation. The resulting homogeneous cosmology yields the closed algebraic equation

$$E^2 = \Omega_{m0} a^{-3} + \Omega_{r0} a^{-4} + \Omega_{R0} E^{2-\theta},$$

together with a proof of existence and uniqueness of the expanding branch, exact expressions for the logarithmic Hubble derivative, deceleration parameter, effective equation of state, acceleration–capacity identity, early- and late-time asymptotics, and a falsifiable standard-siren distance relation with $c_T = 1$.

The RFG-R extension introduced here is a separate, explicitly defined low-energy multiscale EFT. It regularizes the $X \to 0$ action, preserves the original homogeneous branch to controlled order, and matches exactly to GR in a specified local-curvature domain. It consequently gives a definite local PPN prediction. Its regularized action, potential reconstruction, numerical validation, and Cassini likelihood factor are supplied as executable source files. For RFG-R coupled to one minimally coupled canonical scalar, the sourced lapse–shift constraints and the reduced quadratic action are also evaluated exactly, including the GR normalization check $K = 6$, $c_s^2 = 1$. The action is mapped exactly into the extended EFT basis; the map forces a nonzero $\bar{m}_5\, \delta^{(3)} R\, \delta K$ operator that is absent from the public stock H-EFTCAMB input basis [10]. The complete *pure-gravity* extended-EFT scalar action is degenerate. The sourced photon–baryon–CDM–neutrino action, untruncated kinetic hierarchy, and the remaining spatial metric equations are derived directly from the same action; their exact DAE closure has a finite-wavenumber curvature-constraint singularity on the published reference branch. A distinct, background-null RFG-RΞ action is then derived directly from the preferred-foliation invariants ${}^{(3)}R + \sigma_{ij}\sigma^{ij}$: it removes that detected zero on the stated finite DAE grid while leaving the R-Universe Friedmann branch and $c_T = 1$ unchanged. An external GR/CAMB spectrum and Planck low-ell+lensing reference has been executed only to validate the data interface. Neither RFG-R nor RFG-RΞ has an executed cosmological data likelihood.

For the same R-Universe background law, this paper also supplies an explicitly declared covariant Kinetic Gravity Braiding (KGB) realization. Its action reconstruction, global scalar-stability tests, and native photon–baryon–CDM–neutrino H–EFTCAMB evolution are executable. At one fully declared parameter point, the official Planck 2018 TTTEEE, low-ℓ, and lensing likelihood objects give $-2\ln\mathcal{L} = 1210.1599180$. An executed fixed-input extension combines Planck with Pantheon+, DESI DR2 BAO, and chronometers; at the selected matched point, its KGB-minus-ΛCDM conditional sum is -2.5637653. The same KGB point has a direct native $f\sigma_8$ audit and a local screening gate. Four independent exact chains also yield a converged Planck–Pantheon+–DESI DR2 BAO–chronometer posterior for this stated KGB action after the declared 25% per-chain burn-in: the largest rank-normalized folded split-$\hat{R}$ is 1.0066, and the minimum bulk and tail ESS values are 690.2 and 1214.6. A separate local ΛCDM control posterior using the same stated CMB and late-time probe family reaches four independent 12,000-sample chains and recovers a standard benchmark region, with $\Omega_{m0} = 0.2971 \pm 0.0035$, $\omega_b = 0.02240 \pm 0.00012$, $n_s = 0.9682 \pm 0.0033$, $\tau = 0.0386 \pm 0.0086$, $A_{\rm Planck} = 0.9993 \pm 0.0025$, and a best sampled $\chi^2 = 2454.7832$. This control run supports the data-interface and calibration status of the R-Universe analysis, but it is not imposed as a target surface: R-Universe KGB need not copy the individual ΛCDM parameter values. Finally, dedicated dynamic nested-sampling evidence calculations over the explicitly declared Cobaya priors give $\log Z_{\rm KGB} = -1335.5996905 \pm 2.8042019$ and $\log Z_{\Lambda\rm CDM} = -1291.3199169 \pm 2.8135177$, hence $\Delta \log Z = \log Z_{\rm KGB} - \log Z_{\Lambda\rm CDM} = -44.2797735$. Thus the production marginal likelihood favors the local ΛCDM reference for this exact probe set and prior volume, even though the fixed-input conditional profile is slightly better for the selected KGB point. The KGB action remains a physical R-Universe replacement candidate with a completed posterior and evidence audit, but final empirical replacement is not asserted; the RSD compilation is still not a survey likelihood.

Keywords: Relational Capacity Dynamics (RCD); Relational Foliation Gravity (RFG); emergent time; relational capacity; preferred foliation; scalar field theory; modified gravity; Hamiltonian formulation; Noether currents; hyperbolic field equations; cosmology; gravitational waves; standard sirens; Kinetic Gravity Braiding; CMB likelihood.

1 Introduction

A fundamental theory must specify its dynamical variables, symmetries, action, variational equations, physical degrees of freedom, initial-value structure, and falsification conditions. A background equation alone is not sufficient. The purpose of this manuscript is to assemble the mathematically derived RCD and RFG layers together with a deliberately explicit, regular low-energy extension, RFG-R. The distinction is essential: RFG-R is a model completion and matching prescription, not a retroactive theorem about the original singular RFG branch.

We distinguish three logically separate statements that must not be conflated:

(i) **Mathematical definition:** the action and its variational problem are specified.

(ii) **Internal viability:** all propagating modes must be ghost-free, gradient-stable, and compatible with a well-posed initial-value problem.

(iii) **Empirical viability:** cosmological, Solar-System, binary-pulsar, gravitational-wave and laboratory data must be fitted with a reproducible likelihood.

Replacement criterion. The replacement claim tested here is stronger than a request to improve one numerical score against ΛCDM. The standard ΛCDM model is an exceptionally successful phenomenological reference: it supplies a compact background expansion, a calibrated perturbation pipeline, and an economical description of the late Universe in terms of cold dark matter and a cosmological constant. Its success is precisely why it is used throughout this manuscript as the control model. A physical replacement, however, must do more than fit the same background curve. It must identify the underlying variables, state the action principle, derive the field equations, specify the constraint algebra and propagating degrees of freedom, recover local GR where required, propagate photons, baryons, CDM and neutrinos through a reproducible data interface, and expose the points at which the theory can fail. In this sense the R-Universe programme is not presented as a cosmetic reparametrization of ΛCDM. It is presented as an attempt to replace a phenomenological dark-sector description by a structured dynamical account whose ontology, variational equations, stability conditions, tensor sector, local limit, matter interface, and likelihood products are all explicitly auditable.

The manuscript therefore separates two different standards of success. The first is explanatory closure: how much of the cosmological framework follows from specified fields, symmetries, actions, constraints and matching rules, rather than being inserted as an effective component. The second is empirical promotion: whether the resulting completed model wins, or at least survives, a matched multi-probe likelihood and Bayesian-evidence comparison under declared priors. The present work substantially advances the first standard and makes the second standard reproducible for the covariant KGB realization. It does not hide the empirical outcome: the production evidence reported below favors the local ΛCDM reference for the stated KGB comparison. Thus the manuscript argues for a deeper replacement candidate, not for an already completed empirical replacement of ΛCDM.

The original calculation completes item (i) for the scalar macroscopic sector and for the homogeneous and tensor sectors of the preferred-foliation theory. RFG-R additionally defines a regular action and an exact local-GR domain, but it does not make the original scalar perturbation degeneracy disappear. Preferred-foliation and ADM-based constructions provide the broader mathematical context [3, 4, 5, 6]. Its ADM action is now mapped to the extended EFT basis [12]; that mapping identifies the operator which a full 3+1 Einstein–Boltzmann implementation must retain, rather than replacing the perturbation theory by a background-only surrogate.

The final empirical section adds an action-faithful data execution without rewriting the RFG-R result. It gives a covariant KGB realization of the same R-Universe background law, reconstructs its four scalar functions, and evolves that action through the reparametrized-Horndeski interface of H–EFTCAMB. This is an additional realization within the present manuscript, not an identification of the KGB action with RFG-R: the fixed-point likelihood, conditional multi-probe record, and exact posterior reported below apply to the stated KGB action alone.

The logical hierarchy followed throughout is

$$\text{ontology} \longrightarrow \text{state space} \longrightarrow \text{action} \longrightarrow \text{equations} \longrightarrow \text{stability} \longrightarrow \text{observables} \longrightarrow \text{likelihoods} \longrightarrow \text{fa}$$

Table 1: Scope comparison for the constructions in this paper. "Defined" is not a data fit, and an entry for a particular sector does not imply health of every perturbative sector.

Framework	Scalar status	Tensor speed	Effective $w_R(a)$	Analytic background closure
ΛCDM	N/A	$c_T = 1$	$w = -1$ (const.)	Yes
Quintessence / K-essence	Conditional ($c_s^2 > 0$)	$c_T = 1$	Potential-dependent	Selected potentials only
Horndeski / DHOST	After cleaning	Generally $c_T(z) \neq 1$	Dynamical	Varies
RCD / original RFG	RCD stable; RFG FLRW scalar degeneracy	$c_T = 1$ (exact)	$w_R(a) > -1$ (derived)	Yes (Eq. (21))
RFG-R EFT	Exact finite multi-fluid constraint core; kinetic hierarchy	$c_T = 1$ on FLRW	Regularized branch	Yes (Eq. (31))
Covariant KGB R-Universe	Positive Q_s, c_s^2 on stated scan	$c_T = 1$ (exact)	Reconstructed	Yes (Eq. (171))

Table 2: Closure ledger used throughout the manuscript. The table prevents a common failure mode in speculative theory papers: treating a selected branch, reconstruction condition, or calculational target as if it were already a theorem.

Layer	Status in this manuscript	Precise content
RCD scalar ontology and action	Closed classical sector	$\mathcal{C} > 0$, $\varphi = \ln(\mathcal{C}/\mathcal{C}_\star)$, action, Euler–Lagrange equation, Hamiltonian, Noether currents, linear hyperbolicity.
Finite-response transport	Closed once constitutive data are fixed	Maxwell–Cattaneo relaxation law and entropy-production condition; not a fundamental microscopic derivation.
RCD–RFG identification	Branch postulate	The map from $\mathcal{C}$ to the homogeneous $X = K/(3H_0)$ branch is imposed through the capacity-scaling law, not yet derived from a common microscopic action.
RFG homogeneous background	Derived after branch postulate	$Q(X)$, $V(X)$, Friedmann constraint, uniqueness of $E(a)$, $w_R(a)$, $q(a)$, and asymptotic limits.
RFG tensor sector	Derived within the action	Q_T, $c_T = 1$, no tensor ghost on the branch, and the standard-siren amplitude ratio.
RFG pure-gravity FLRW scalar reduction	Computed	The lapse and scalar shift are solved algebraically; the exact reduced density is $L_{\rm red} = A\zeta\dot{\zeta} + B\zeta^2$, with no $\dot{\zeta}^2$ term. After integration by parts, $S^{(2)}_{\rm red}$ is algebraic in time.
RFG full scalar constraints with matter	Open completion gate	The metric Legendre rank and the local $\mathcal{C}$–$\mathcal{C}$ bracket are explicit. The global kernel, boundary-condition dependence, complete constraint classification, and matter-coupled scalar reduction remain open before a final degree-of-freedom statement can be claimed.
RFG-R local gravitational domain	Defined EFT matching sector	A smooth Weyl-invariant switch makes the local action and all its variations exactly GR; this yields $\gamma = \beta = 1$, $\alpha_1 = \alpha_2 = 0$ wherever the switch is constant.
RFG-R canonical scalar	Computed reference-sector reduction	For one minimally coupled canonical scalar in comoving gauge, the sourced lapse–shift Hessian, its determinant, and the exact Schur-complement action are derived. The GR stiff-field limit and a regular reference branch pass the stated kinetic and low-gradient checks.

Table 3: Closure ledger used throughout the manuscript (continued).

Layer	Status in this manuscript	Precise content
RFG-R multi-fluid core	Computed action, hierarchy, and DAE closure	Exact Sorkin–Schutz constraint matrix, full $\bar{m}_5$ contribution, spatial metric equations, rational GR rank check, documented RFG-R inertia audit, and a derived finite-k curvature-constraint singularity.
RFG-RΞ completion	New action; finite DAE gate and local data factor passed	The background-null $\Xi({}^{(3)}R + \sigma_{ij}\sigma^{ij})$ operator is directly varied. At $\Xi = 1$ and $\Xi = 2$, separate 49-by-81 (a, k) audits have no sampled μ_ζ root; the local Cassini factor is computed exactly in the GR-matched domain.
Matter/CMB likelihood	GR data-interface reference executed; no R-Universe cosmological likelihood	CAMB 2.0.1 and official Planck low-ell+lensing components run at a fixed GR point. The original RFG-R DAE coefficient μ_ζ crosses zero, while RFG-RΞ has not been through an action-faithful Boltzmann solver or joint CMB/matter likelihood.
Covariant KGB likelihood and evidence	Fixed point, conditional CMB–late-time record, exact posterior, production evidence, and native RSD audit executed	The separately defined KGB realization is evolved with photons, baryons, CDM, massless neutrinos, recombination, and lensing in native H–EFTCAMB. A fixed-input Planck–Pantheon+–DESI DR2 BAO–chronometer record, a converged exact posterior for that stated probe set, a local screening gate, and a dedicated dynamic nested-sampling evidence calculation are executed. The matched local ΛCDM reference evidence is also executed. The resulting $\Delta \log Z = -44.2797735$ favors the local ΛCDM reference over KGB for the declared priors and probes; RSD is kept as a numerical residual audit because its survey likelihood inputs are unavailable.

2 Primitive ontology and state space

Definition 2.1 (Configuration ordering)**.** *The parameter $\tau \in I \subseteq \mathbb{R}$ orders macroscopic configurations. It is not assumed to coincide with physical clock time.*

Definition 2.2 (Spatial domain)**.** *Let $\Omega \subseteq \mathbb{R}^3$ be a connected spatial domain with coordinates $\mathbf{x} = (x^1, x^2, x^3)$. The Euclidean operators ∇, $\nabla\cdot$ and ∇^2 are kinematic structures of the present* *continuum realization.*

Definition 2.3 (Relational capacity)**.** *The primitive macroscopic scalar field is*

$$\mathcal{C} : I \times \Omega \to (0, \infty).$$

Its value measures the locally available capacity of the underlying physical substrate to support distinguishable relations and dynamical change.

Strict positivity $\mathcal{C} > 0$ permits the dimensionless ratio

$$\chi := \frac{\mathcal{C}}{\mathcal{C}_\star} > 0, \qquad \varphi := \ln \chi = \ln \frac{\mathcal{C}}{\mathcal{C}_\star},$$

where $\mathcal{C}_\star$ is a fixed reference capacity of the same dimension as $\mathcal{C}$. The inverse relation is $\mathcal{C} = \mathcal{C}_\star \, \mathrm{e}^{\varphi}$.

Exact differential identities follow at once:

$$\partial_\tau \mathcal{C} = \mathcal{C} \, \partial_\tau \varphi, \tag{1}$$
$$\nabla \mathcal{C} = \mathcal{C} \, \nabla \varphi, \tag{2}$$
$$\nabla^2 \mathcal{C} = \mathcal{C}\big(\nabla^2 \varphi + |\nabla \varphi|^2\big). \tag{3}$$

Definition 2.4 (Capacity current and flow velocity)**.** *The vector field $\mathbf{J}_\mathcal{C} : I \times \Omega \to \mathbb{R}^3$ is the flux of relational capacity. Where $\mathcal{C} > 0$,*

$$\mathbf{A}_\mathcal{C} := \frac{\mathbf{J}_\mathcal{C}}{\mathcal{C}} \qquad \Rightarrow \qquad \mathbf{J}_\mathcal{C} = \mathcal{C} \mathbf{A}_\mathcal{C}.$$

3 Dimensional architecture

Independent base dimensions are $[L]$, $[T]$, $[M]$ and, provisionally, $[K]$ (length, physical time, mass, capacity). The ordering parameter is assigned $[\tau] = T$.

Quantity	Definition	Dimension
τ, t	ordering / physical time	T
$\mathbf{x}$	spatial label	L
$\mathcal{C}, \mathcal{C}_\star$	capacity	K
χ, φ	$\mathcal{C}/\mathcal{C}_\star$, $\ln \chi$	1
$\mathbf{J}_\mathcal{C}$	capacity current	KLT^{-1}
$\mathbf{A}_\mathcal{C}$	$\mathbf{J}_\mathcal{C}/\mathcal{C}$	LT^{-1}
$\mathcal{S}_\mathcal{C}$	capacity source	KT^{-1}
$\rho_\mathcal{C}$	kinetic coefficient	ML^{-1}
$\kappa_\mathcal{C}$	gradient coefficient	MLT^{-2}
U	potential density	$ML^{-1}T^{-2}$
Π_φ	canonical momentum density	$ML^{-1}T^{-1}$
$\mathcal{H}$	Hamiltonian density	$ML^{-1}T^{-2}$
γ	clock exponent	1
$c_\mathcal{C}$	$\sqrt{\kappa_\mathcal{C}/\rho_\mathcal{C}}$	LT^{-1}

The characteristic speed of the scalar sector is

$$c_\mathcal{C} := \sqrt{\frac{\kappa_\mathcal{C}}{\rho_\mathcal{C}}}.$$

Remark 3.1 (Status of the capacity dimension)**.** *The independent dimension K is introduced solely to keep the ontology of the capacity field $\mathcal{C}$ and its current $\mathbf{J}_{\mathcal{C}}$ dimensionally consistent with the balance law $\partial_\tau \mathcal{C} + \nabla \cdot \mathbf{J}_{\mathcal{C}} = \mathcal{S}_{\mathcal{C}}$. In the conservative scalar action of the present paper the dynamics depend only on the dimensionless variable $\varphi = \ln(\mathcal{C}/\mathcal{C}_\star)$. Consequently the coefficients $\rho_{\mathcal{C}}$ and $\kappa_{\mathcal{C}}$ carry no residual factor of K. The dimension K becomes dynamically relevant only when the capacity current is retained as an independent degree of freedom or when matter couples non-minimally to $\mathcal{C}$ itself. Until those sectors are constructed, K remains a bookkeeping device that does not affect the equations derived below.*

4 Fundamental postulates

Axiom 4.1 (Relational primacy)**.** *The macroscopic physical state contains a positive relational-capacity field $\mathcal{C}$. Observable local rates may depend on $\mathcal{C}$ and its derivatives.*

Axiom 4.2 (Variational dynamics)**.** *The reversible sector follows from a stationary action under compactly supported variations or under variations satisfying explicitly stated boundary conditions.*

Axiom 4.3 (Capacity balance)**.**

$$\partial_\tau \mathcal{C} + \nabla \cdot \mathbf{J}_{\mathcal{C}} = \mathcal{S}_{\mathcal{C}}.$$

Axiom 4.4 (Clock emergence)**.** *Physical clocks measure a time t related to the ordering parameter by*

$$\mathrm{d}t = \chi^\gamma \mathrm{d}\tau,$$

where $\gamma \in \mathbb{R}$ is universal within a chosen phase of the theory.

Axiom 4.5 (Locality of the macroscopic conservative sector)**.** *At lowest derivative order the conservative scalar action depends on φ, $\partial_\tau \varphi$ and $\nabla \varphi$ only.*

5 Conservative scalar action: complete variational calculation

The minimal conservative scalar action is

$$S_{\mathcal{C}}[\varphi] = \int_{\tau_1}^{\tau_2} \mathrm{d}\tau \int_\Omega \mathrm{d}^3x \, \mathcal{L}_{\mathcal{C}}, \tag{4}$$

with

$$\mathcal{L}_{\mathcal{C}} = \frac{\rho_{\mathcal{C}}}{2}(\partial_\tau \varphi)^2 - \frac{\kappa_{\mathcal{C}}}{2}|\nabla \varphi|^2 - U(\varphi).$$

The measure $\mathrm{d}\tau \, \mathrm{d}^3x$ has dimension TL^3 and the Lagrangian density has dimension $ML^{-1}T^{-2}$, so the action has the standard dimension ML^2T^{-1}.

Let $\varphi_\epsilon = \varphi + \epsilon\eta$, where η is smooth and either compactly supported or satisfies boundary conditions that make all boundary terms vanish. Differentiating at $\epsilon = 0$ yields

$$\delta S_{\mathcal{C}} = \int \mathrm{d}\tau \, \mathrm{d}^3x \Big[\rho_{\mathcal{C}} \, (\partial_\tau \varphi)(\partial_\tau \eta) - \kappa_{\mathcal{C}} \nabla \varphi \cdot \nabla \eta - U_{,\varphi}\eta\Big]. \tag{5}$$

The temporal kinetic term integrates by parts to

$$\int_{\tau_1}^{\tau_2} \mathrm{d}\tau \, \rho_{\mathcal{C}} \, (\partial_\tau \varphi)(\partial_\tau \eta) = \Big[\rho_{\mathcal{C}} \, (\partial_\tau \varphi)\eta\Big]_{\tau_1}^{\tau_2} - \int_{\tau_1}^{\tau_2} \mathrm{d}\tau \, \rho_{\mathcal{C}} \, (\partial_\tau^2 \varphi)\eta. \tag{6}$$

The endpoint contribution vanishes by the temporal boundary condition.

The spatial-gradient term integrates by parts to

$$-\int_{\Omega} \mathrm{d}^3x\, \kappa_{\mathcal{C}} \nabla\varphi \cdot \nabla\eta = -\int_{\partial\Omega} \mathrm{d}A\, \kappa_{\mathcal{C}}\, (\mathbf{n}\cdot\nabla\varphi)\eta + \int_{\Omega} \mathrm{d}^3x\, \kappa_{\mathcal{C}}\, (\nabla^2\varphi)\eta. \tag{7}$$

Collecting terms,

$$\delta S_{\mathcal{C}} = \text{boundary terms} + \int \mathrm{d}\tau\, \mathrm{d}^3x \Big[-\rho_{\mathcal{C}}\, \partial_\tau^2\varphi + \kappa_{\mathcal{C}}\nabla^2\varphi - U_{,\varphi}\Big]\eta. \tag{8}$$

Theorem 5.1 (Scalar field equation)**.** *Stationarity of $S_{\mathcal{C}}$ under admissible variations yields*

$$\boxed{\rho_{\mathcal{C}}\, \partial_\tau^2\varphi - \kappa_{\mathcal{C}}\nabla^2\varphi + U_{,\varphi} = 0.}$$

In terms of $\mathcal{C} = \mathcal{C}_\star\, \mathrm{e}^{\varphi}$ the equation becomes

$$\rho_{\mathcal{C}}\left(\frac{\partial_\tau^2\mathcal{C}}{\mathcal{C}} - \frac{(\partial_\tau\mathcal{C})^2}{\mathcal{C}^2}\right) - \kappa_{\mathcal{C}}\left(\frac{\nabla^2\mathcal{C}}{\mathcal{C}} - \frac{|\nabla\mathcal{C}|^2}{\mathcal{C}^2}\right) + U_{,\varphi}\Big(\ln\frac{\mathcal{C}}{\mathcal{C}_\star}\Big) = 0. \tag{9}$$

6 Hamiltonian formulation

Definition 6.1 (Canonical momentum)**.**

$$\Pi_\varphi := \frac{\partial\mathcal{L}_{\mathcal{C}}}{\partial(\partial_\tau\varphi)} = \rho_{\mathcal{C}}\, \partial_\tau\varphi.$$

The Hamiltonian density obtained by Legendre transform is

$$\boxed{\mathcal{H}_{\mathcal{C}} = \frac{\Pi_\varphi^2}{2\rho_{\mathcal{C}}} + \frac{\kappa_{\mathcal{C}}}{2}|\nabla\varphi|^2 + U(\varphi).}$$

Theorem 6.2 (Hamiltonian equivalence)**.** *For $\rho_{\mathcal{C}} \neq 0$ and admissible boundary conditions the Hamilton equations are equivalent to the Euler–Lagrange equation.*

Theorem 6.3 (Energy boundedness)**.** *If Ω has finite volume, $\rho_{\mathcal{C}} > 0$, $\kappa_{\mathcal{C}} > 0$, $U(\varphi) \geq U_{\min} > -\infty$, and $\mathcal{H}_{\mathcal{C}}$ is integrable, then*

$$H_{\mathcal{C}} := \int_{\Omega} \mathcal{H}_{\mathcal{C}}\, \mathrm{d}^3x \geq U_{\min}\, \mathrm{Vol}(\Omega).$$

7 Noether identities with local currents

7.1 Energy

Time-translation invariance of the Lagrangian density yields the local energy conservation law

$$\boxed{\partial_\tau\mathcal{E} + \nabla\cdot\mathbf{S}_\varphi = 0,}$$

where the energy density and flux are

$$\mathcal{E} = \frac{\rho_{\mathcal{C}}}{2}(\partial_\tau\varphi)^2 + \frac{\kappa_{\mathcal{C}}}{2}|\nabla\varphi|^2 + U(\varphi), \tag{10}$$

$$\mathbf{S}_\varphi = -\kappa_{\mathcal{C}}\, (\partial_\tau\varphi)\nabla\varphi. \tag{11}$$

7.2 Momentum

Spatial translation invariance yields the local momentum conservation law

$$\boxed{\partial_\tau \mathcal{P}_i - \partial_j \Sigma_{ij} = 0,}$$

with momentum density $\mathcal{P}_i = \rho_{\mathcal{C}}\,(\partial_\tau\varphi)\,(\partial_i\varphi)$ and a convenient stress tensor

$$\Sigma_{ij} = \kappa_{\mathcal{C}}\,(\partial_i\varphi)(\partial_j\varphi) + \delta_{ij}\left[\frac{\rho_{\mathcal{C}}}{2}(\partial_\tau\varphi)^2 - \frac{\kappa_{\mathcal{C}}}{2}|\nabla\varphi|^2 - U(\varphi)\right]. \tag{12}$$

8 Linearization, dispersion and stability

Around a static vacuum φ_0 with $U_{,\varphi}(\varphi_0) = 0$ the quadratic action for the perturbation ψ yields the dispersion relation

$$\omega^2 = c_{\mathcal{C}}^2 k^2 + \omega_0^2, \qquad c_{\mathcal{C}}^2 = \frac{\kappa_{\mathcal{C}}}{\rho_{\mathcal{C}}}, \qquad \omega_0^2 = \frac{U_{,\varphi\varphi}(\varphi_0)}{\rho_{\mathcal{C}}}.$$

Theorem 8.1 (Linear scalar viability)**.** *The static vacuum is linearly stable if and only if*

$$\boxed{\rho_{\mathcal{C}} > 0, \qquad \kappa_{\mathcal{C}} > 0, \qquad U_{,\varphi\varphi}(\varphi_0) \geq 0.}$$

The principal symbol is strictly hyperbolic with characteristic speeds $\pm c_{\mathcal{C}}$ whenever $\rho_{\mathcal{C}} > 0$ and $\kappa_{\mathcal{C}} > 0$.

Open problem 8.2 (Global nonlinear well-posedness)**.** *Determine the broadest class of potentials $U(\varphi)$ for which finite-energy solutions exist globally.*

9 Finite-response transport

Definition 9.1 (Capacity free energy and chemical potential)**.** *For a local capacity free-energy functional*

$$\mathfrak{F}[\mathcal{C}] = \int_\Omega \left[f(\mathcal{C}) + \frac{\ell_{\mathcal{C}}^2}{2}|\nabla\mathcal{C}|^2\right] \mathrm{d}^3x,$$

the chemical potential is

$$\mu_{\mathcal{C}} := \frac{\delta\mathfrak{F}}{\delta\mathcal{C}} = f_{,\mathcal{C}} - \ell_{\mathcal{C}}^2 \nabla^2\mathcal{C}.$$

The mobility $\mathcal{M} > 0$ relates current to chemical-potential gradient, and $\tau_J > 0$ is the finite current-relaxation time.

Axiom 9.2 (Finite-response constitutive law)**.**

$$\tau_J \partial_\tau \mathbf{J}_{\mathcal{C}} + \mathbf{J}_{\mathcal{C}} = -\mathcal{M}\nabla\mu_{\mathcal{C}}, \qquad \partial_\tau\mathcal{C} + \nabla\cdot\mathbf{J}_{\mathcal{C}} = 0.$$

Taking the divergence of the first equation and using the balance law gives

$$\boxed{\tau_J \partial_\tau^2 \mathcal{C} + \partial_\tau\mathcal{C} = \mathcal{M}\nabla^2\mu_{\mathcal{C}}.}$$

For perturbations around a homogeneous state $\mathcal{C}_0$, and at wavelengths much longer than $\ell_{\mathcal{C}}$,

$$\delta\mu_{\mathcal{C}} = f_{,\mathcal{C}\mathcal{C}}(\mathcal{C}_0)\delta\mathcal{C}.$$

Defining

$$c_J^2 := \frac{\mathcal{M} f_{,\mathcal{CC}}(\mathcal{C}_0)}{\tau_J},$$

one obtains

$$\partial_\tau^2 \delta\mathcal{C} + \frac{1}{\tau_J}\partial_\tau \delta\mathcal{C} - c_J^2 \nabla^2 \delta\mathcal{C} = 0.$$

For

$$\delta\mathcal{C} \propto \exp[i(\mathbf{k}\cdot\mathbf{x} - \omega\tau)]\,,$$

$$\omega_\pm = -\frac{i}{2\tau_J} \pm \sqrt{c_J^2 k^2 - \frac{1}{4\tau_J^2}}.$$

Propagation occurs for

$$k > \frac{1}{2c_J\tau_J},$$

whereas lower wave numbers are overdamped. Local thermodynamic stability requires $f_{,\mathcal{CC}}(\mathcal{C}_0) > 0$, which implies $c_J^2 > 0$ when $\mathcal{M} > 0$ and $\tau_J > 0$.

10 Emergent physical time

The clock map $\mathrm{d}t = \chi^\gamma \mathrm{d}\tau$ implies the exact conversion rules

$$\partial_\tau = \chi^\gamma \partial_t, \tag{13}$$

$$\partial_\tau^2 F = \chi^{2\gamma}\Big[\partial_t^2 F + \gamma(\partial_t\varphi)(\partial_t F)\Big]. \tag{14}$$

In particular,

$$\partial_\tau^2 \varphi = \chi^{2\gamma}\Big[\partial_t^2 \varphi + \gamma(\partial_t\varphi)^2\Big].$$

Remark 10.1 (Integrability). *When $\chi = \chi(t,\mathbf{x})$ varies spatially, the relation $\mathrm{d}t = \chi^\gamma \mathrm{d}\tau$ is not a global exact differential. Equal increments of τ then correspond to position-dependent physical clock increments. This fact must be taken into account in any future inhomogeneous or particle analysis.*

11 Covariant completion: Relational Foliation Gravity

11.1 Postulates

Axiom 11.1 (Spacetime and matter). *Spacetime is a four-dimensional Lorentzian manifold with metric $g_{\mu\nu}$ of signature $(-+++)$. Matter fields couple minimally and universally to $g_{\mu\nu}$, so that $\nabla_\mu T^{\mu\nu}_{(m)} = 0$.*

Axiom 11.2 (Physical foliation). *A scalar khronon field $T(x)$ with timelike gradient defines a physical ordering of hypersurfaces. The future-directed unit normal is*

$$u_\mu = -\frac{\nabla_\mu T}{\sqrt{-g^{\alpha\beta}\nabla_\alpha T \nabla_\beta T}}.$$

The theory is invariant under diffeomorphisms and strictly increasing monotonic reparameterizations

$$T \mapsto f(T), \qquad f'(T) > 0,$$

so that the future orientation of u_μ is preserved. Orientation-reversing maps would send $u_\mu \to -u_\mu$ and $K \to -K$; for noninteger powers such as $X^{-\theta}$ and $X^{2-\theta}$ this would not define a single real branch. They are therefore not gauge redundancies of the branch studied here.

Axiom 11.3 (Relational capacity on the homogeneous branch)**.**

$$\Omega_R(X) = \Omega_{R0} X^{-\theta}, \qquad X \equiv \frac{K}{3H_0}, \qquad K = \nabla_\mu u^\mu,$$

with $\theta > 0$.

The connection between the macroscopic RCD capacity field $\mathcal{C}$ and the homogeneous RFG branch variable X is presently fixed by this homogeneous capacity-scaling law. It is not yet derived from a common microscopic action. This statement is part of the theory ledger: all equations below follow from the RFG action once the branch law $\Omega_R(X)$ is specified, but the deeper RCD–RFG identification remains a separate completion problem.

Axiom 11.4 (Paired geometric response)**.** *The same capacity that modifies the homogeneous Hamiltonian renormalizes, in equal measure, the kinetic and spatial-gradient terms of the transverse-traceless metric perturbations, thereby enforcing* $c_T = 1$*.*

11.2 Fundamental action

The four-dimensional foliation action is

$$\boxed{S_g = \frac{M_{\rm Pl}^2}{2}\int \mathrm{d}^4x\,\sqrt{-g}\,\Big[Q(X)\big({}^{(3)}R[h,u] + K_{\mu\nu}K^{\mu\nu} - K^2\big) + 2H_0^2 V(X)\Big],} \tag{15}$$

where

$$h_{\mu\nu} = g_{\mu\nu} + u_\mu u_\nu, \qquad K_{\mu\nu} = h_\mu{}^\alpha h_\nu{}^\beta \nabla_\alpha u_\beta, \qquad K = \nabla_\mu u^\mu, \qquad X = \frac{K}{3H_0}.$$

The scalar ${}^{(3)}R[h,u]$ is the Ricci scalar of the hypersurfaces orthogonal to u^μ. Equation (15) is invariant under four-dimensional diffeomorphisms and under the orientation-preserving reparameterizations of T stated above, because it depends on T only through the unit normal u_μ.

In ADM variables adapted to the physical foliation, Eq. (15) becomes

$$\boxed{S = S_g + S_m, \quad S_g = \frac{M_{\rm Pl}^2}{2}\int \mathrm{d}t\,\mathrm{d}^3x\, N\sqrt{\gamma}\Big[Q(X)\big({}^{(3)}R + K_{ij}K^{ij} - K^2\big) + 2H_0^2\,V(X)\Big].} \tag{16}$$

Here ${}^{(3)}R$ has dimension L^{-2} and the extrinsic-curvature combination $K_{ij}K^{ij} - K^2$ has dimension T^{-2}. With units in which $c = 1$ these dimensions coincide. The factor H_0^2 multiplies the dimensionless potential $V(X)$ so that $H_0^2 V(X)$ carries the same dimension as the curvature terms. The two free functions are fixed by the tensor-normalization and background-reconstruction requirements:

$$\boxed{Q(X) = 1 - AX^{-\theta}, \qquad A = \frac{\Omega_{R0}}{1+\theta},} \tag{17}$$

$$\boxed{V(X) = \frac{6\theta\Omega_{R0}}{(1+\theta)(1-\theta)} X^{2-\theta} \qquad (\theta \neq 1).} \tag{18}$$

For the singular case $\theta = 1$ the reconstruction equation admits the continuous limit

$$\boxed{V_{\theta=1}(X) = 3\Omega_{R0}\, X \ln X} \tag{19}$$

(after a pure boundary term linear in X has been discarded). Equation (19) is therefore the unique regular extension of Eq. (18) through $\theta = 1$.

11.3 FLRW reduction and reconstruction of the free functions

For

$$\mathrm{d}s^2 = -N^2(t)\mathrm{d}t^2 + a^2(t)\delta_{ij}\mathrm{d}x^i\mathrm{d}x^j,$$

the shift vanishes and

$$K_{ij} = H_N\gamma_{ij}, \qquad K = 3H_N, \qquad H_N := \frac{\dot a}{aN}, \qquad X = \frac{H_N}{H_0}.$$

Hence

$${}^{(3)}R = 0, \qquad K_{ij}K^{ij} - K^2 = -6H_N^2.$$

After removing the constant comoving volume,

$$L_g = M_{\rm Pl}^2 H_0^2 N a^3 F(X), \qquad F(X) := -3X^2 Q(X) + V(X).$$

Since $X \propto N^{-1}$,

$$\frac{\partial X}{\partial N} = -\frac{X}{N}.$$

The lapse variation gives

$$M_{\rm Pl}^2 H_0^2 [F - XF_{,X}] = \rho,$$

or

$$X^2 Q + X^3 Q_{,X} + \frac{V - XV_{,X}}{3} = \frac{\rho}{3M_{\rm Pl}^2 H_0^2}. \tag{20}$$

Theorem 11.5 (Reconstruction of V)**.** *Assume*

$$Q(X) = 1 - AX^{-\theta}, \qquad A = \frac{\Omega_{R0}}{1+\theta},$$

and require

$$X^2 = \frac{\rho}{3M_{\rm Pl}^2 H_0^2} + \Omega_{R0}X^{2-\theta}.$$

Then, up to a term linear in X,

$$V(X) = \frac{6\theta\Omega_{R0}}{(1+\theta)(1-\theta)}X^{2-\theta}, \qquad \theta \neq 1.$$

Proof. Because

$$Q_{,X} = A\theta X^{-\theta-1},$$
$$X^2 Q + X^3 Q_{,X} = X^2 + A(\theta - 1)X^{2-\theta}.$$

Comparison with the required background equation yields

$$V - XV_{,X} = -\frac{6\theta\Omega_{R0}}{1+\theta}X^{2-\theta}.$$

Writing $V = XY$ gives

$$-X^2 Y_{,X} = -\frac{6\theta\Omega_{R0}}{1+\theta}X^{2-\theta},$$

and therefore

$$Y_{,X} = \frac{6\theta\Omega_{R0}}{1+\theta}X^{-\theta}.$$

Integration gives

$$V(X) = \frac{6\theta\Omega_{R0}}{(1+\theta)(1-\theta)}X^{2-\theta} + CX.$$

The term CX is the homogeneous solution and is a boundary contribution on the homogeneous branch. □

For $\theta = 1$,

$$V - XV_{,X} = -3\Omega_{R0}X.$$

With $V = XY$,

$$Y_{,X} = \frac{3\Omega_{R0}}{X},$$

so

$$V_{\theta=1}(X) = 3\Omega_{R0}X \ln X + CX.$$

Discarding CX reproduces Eq. (19).

11.4 Quadratic tensor reduction

Use

$$\gamma_{ij} = a^2(t)(e^h)_{ij}, \qquad \partial_i h_{ij} = 0, \qquad h_{ii} = 0.$$

Since

$$\det\left(e^h\right) = e^{\operatorname{tr} h} = 1,$$

one has $\sqrt{\gamma} = a^3$ exactly and

$$K = \frac{1}{N}\partial_t \ln \sqrt{\gamma} = 3H_N.$$

Thus X has no transverse-traceless perturbation. At quadratic order,

$$K_{ij}K^{ij} - K^2 = -6H_N^2 + \frac{1}{4N^2}\dot{h}_{ij}\dot{h}_{ij} + \cdots ,$$

and

$${}^{(3)}R = -\frac{1}{4a^2}\partial_k h_{ij}\partial_k h_{ij} + \cdots ,$$

where omitted terms are background terms or total derivatives. Using the background equations to remove nondynamical quadratic terms,

$$S_T^{(2)} = \frac{M_{\rm Pl}^2}{8}\int \mathrm{d}t\, \mathrm{d}^3x\, a^3 Q(X)\left[\dot{h}_{ij}\dot{h}_{ij} - \frac{\partial_k h_{ij}\partial_k h_{ij}}{a^2}\right].$$

Therefore

$$Q_T = Q(X), \qquad c_T^2 = 1.$$

Since $AX^{-\theta} = \Omega_R/(1+\theta)$,

$$Q_T(a) = 1 - \frac{\Omega_R(a)}{1+\theta}.$$

11.5 Homogeneous cosmology: existence and uniqueness

Definition 11.6 (Dimensionless homogeneous variables)**.**

$$E(a) := \frac{H(a)}{H_0}, \qquad \Omega_m(a) := \frac{\Omega_{m0}a^{-3}}{E^2(a)},$$

$$\Omega_r(a) := \frac{\Omega_{r0}a^{-4}}{E^2(a)}, \qquad \Omega_R(a) := \Omega_{R0}E^{-\theta}(a).$$

The normalization $E(1) = 1$ requires

$$\Omega_{m0} + \Omega_{r0} + \Omega_{R0} = 1.$$

On a flat FLRW background the Hamiltonian constraint reduces to the closed algebraic equation

$$\boxed{E^2(a) = \Omega_{m0}a^{-3} + \Omega_{r0}a^{-4} + \Omega_{R0}E^{2-\theta}(a).} \tag{21}$$

Theorem 11.7 (Unique expanding branch)**.** *For $0 < \theta < 2$ and $0 < \Omega_{R0} < 1$ the equation $f(E;a) = E^2 - \Omega_{R0}E^{2-\theta} - S(a) = 0$, with $S(a) = \Omega_{m0}a^{-3} + \Omega_{r0}a^{-4} > 0$, possesses a unique positive root satisfying $E^\theta > \Omega_{R0}$.*

Proof. Because $E > 0$, Eq. (21) can be divided by $E^{2-\theta}$. The root problem is then equivalent to

$$h(E;a) := E^\theta - \Omega_{R0} - S(a)E^{\theta-2} = 0.$$

For $0 < \theta < 2$,

$$\lim_{E\to 0^+} h(E;a) = -\infty, \qquad \lim_{E\to\infty} h(E;a) = +\infty,$$

because $E^{\theta-2} \to \infty$ at the origin and $E^\theta \to \infty$ at infinity. Therefore at least one positive root exists. Moreover,

$$\frac{\partial h}{\partial E} = \theta E^{\theta-1} + (2-\theta)S(a)E^{\theta-3} > 0$$

for every $E > 0$. Thus $h(E;a)$ is strictly increasing on the whole positive half-line and has exactly one positive zero. At that zero,

$$E^\theta - \Omega_{R0} = S(a)E^{\theta-2} > 0,$$

so $E^\theta > \Omega_{R0}$. This proves both existence and global uniqueness of the expanding branch. □

Dividing Eq. (21) by E^2 yields the exact closure identity

$$\Omega_m(a) + \Omega_r(a) + \Omega_R(a) = 1.$$

Remark 11.8 (Physical meaning of the interval $0 < \theta < 2$)**.** *The endpoints of the allowed interval recover two familiar limits. As $\theta \to 0^+$ one has $E^{2-\theta} \to E^2$, so the relational term simply renormalizes the effective Planck mass (or Newton's constant) and does not drive acceleration by itself. As $\theta \to 2^-$ one has $E^{2-\theta} \to 1$, recovering a pure cosmological-constant contribution. Intermediate values $0 < \theta < 2$ produce a dynamical dark-energy component whose equation-of-state parameter $w_R(a)$ evolves with the expansion. The numerical examples below use a matter density compatible with the standard late-time cosmological reference range [9]. No observational preference for θ is claimed here; a dedicated likelihood analysis is left for future work.*

11.6 Exact logarithmic derivative, deceleration and equation of state

Differentiating Eq. (21) with respect to $N = \ln a$ produces

$$\frac{\mathrm{d}\ln E}{\mathrm{d}\ln a} = -\frac{3\Omega_m(a) + 4\Omega_r(a)}{2 - (2-\theta)\Omega_R(a)}. \tag{22}$$

The deceleration parameter is

$$q = -1 - \frac{\mathrm{d}\ln E}{\mathrm{d}\ln a}.$$

The effective equation of state of the relational sector is derived, not assumed:

$$\boxed{w_R(a) = -1 - \frac{2-\theta}{3}\frac{\mathrm{d}\ln E}{\mathrm{d}\ln a}.} \tag{23}$$

Theorem 11.9 (Non-phantom relational component)**.** *For $0 < \theta < 2$ on the expanding branch,*

$$w_R(a) > -1$$

whenever $\Omega_m + \Omega_r > 0$.

Proof. The numerator $3\Omega_m + 4\Omega_r$ is positive, while

$$2 - (2 - \theta)\Omega_R > \theta > 0.$$

Thus $\mathrm{d}\ln E/\mathrm{d}\ln a < 0$. Since $2 - \theta > 0$,

$$w_R = -1 - \frac{2-\theta}{3}\frac{\mathrm{d}\ln E}{\mathrm{d}\ln a} > -1.$$

□

11.7 Acceleration transition

The exact condition $q = 0$ gives

$$3\Omega_m + 4\Omega_r = 2 - (2 - \theta)\Omega_R.$$

Using $\Omega_m = 1 - \Omega_r - \Omega_R$,

$$\boxed{\Omega_R(z_{\mathrm{acc}}) = \frac{1 + \Omega_r(z_{\mathrm{acc}})}{1 + \theta}.}$$

At the late-time transition $\Omega_r \ll 1$, so

$$\boxed{\Omega_R(z_{\mathrm{acc}}) \simeq \frac{1}{1 + \theta}.}$$

The derived equation of state and the deceleration parameter for three representative values of θ are shown in Fig. 1.

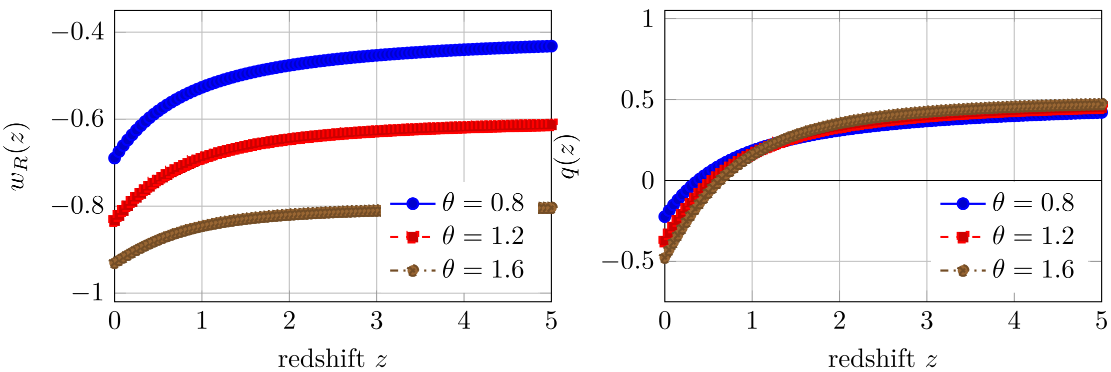

Figure 1: Cosmological dynamics evaluated from the homogeneous RFG equations and rendered from numerical coordinates embedded in this LaTeX source. The left panel shows $w_R(z)$, and the right panel shows $q(z)$, for three representative values of θ. The acceleration transition $q = 0$ coincides with $\Omega_R(z_{\mathrm{acc}}) = 1/(1 + \theta)$. Parameters: $\Omega_{m0} = 0.3$, $\Omega_{r0} = 9 \times 10^{-5}$, and $\Omega_{R0} = 1 - \Omega_{m0} - \Omega_{r0}$.

11.8 Early- and late-time asymptotics

At early times $E \to \infty$, so $\Omega_R = \Omega_{R0}E^{-\theta} \to 0$. The standard radiation- and matter-dominated limits are therefore recovered. The future attractor is de Sitter with

$$E_\infty = \Omega_{R0}^{1/\theta}, \qquad H_\infty = H_0\Omega_{R0}^{1/\theta}.$$

11.9 Tensor sector and standard-siren ratio

The quadratic action for transverse-traceless modes is

$$S_T^{(2)} = \frac{M_{\rm Pl}^2}{8}\int \mathrm{d}t\,\mathrm{d}^3x\, a^3 Q_T\Big[\dot h_{ij}\dot h_{ij} - \frac{\partial_k h_{ij}\partial_k h_{ij}}{a^2}\Big], \tag{24}$$

with

$$\boxed{Q_T(a) = 1 - \frac{\Omega_R(a)}{1+\theta}, \qquad c_T^2 = 1.}$$

The no-ghost condition $Q_T > 0$ holds throughout the homogeneous branch for $\theta > 0$. The standard-siren amplitude relation follows at once:

$$\boxed{\frac{d_L^{\rm GW}(z)}{d_L^{\rm EM}(z)} = \sqrt{\frac{Q_T(0)}{Q_T(z)}}.} \tag{25}$$

Equation (25) is the WKB/subhorizon tensor-amplitude result under the following assumptions: matter is minimally coupled to the metric $g_{\mu\nu}$, photons and gravitational waves share the same background redshift relation, there is no tensor mixing with an additional propagating spin–2 field, and the observed Planck normalization is fixed at $z = 0$. Under these assumptions Q_T modifies only the tensor friction term, not the null cone, which is why $c_T = 1$ and the distance ratio in Eq. (25) can differ from unity simultaneously.

Remark 11.10 (Order-of-magnitude estimate)**.** *For a representative late-time branch with $\Omega_{R0} \approx 0.7$ and $\theta \approx 1.6$ one obtains $Q_T(0) \approx 0.73$. At high redshift $\Omega_R \to 0$, so $Q_T \to 1$ and the ratio approaches $\sqrt{Q_T(0)} \approx 0.85$. At redshift $z = 1$ the same branch yields a ratio of order 0.9. The near-luminal propagation constraint from GW170817 is automatically respected because $c_T = 1$ [7]. The amplitude-distance modification is instead a standard-siren observable of the type discussed in forecast studies [8]. The corresponding curve is shown in Fig. 2. A dedicated likelihood comparison is left for future work.*

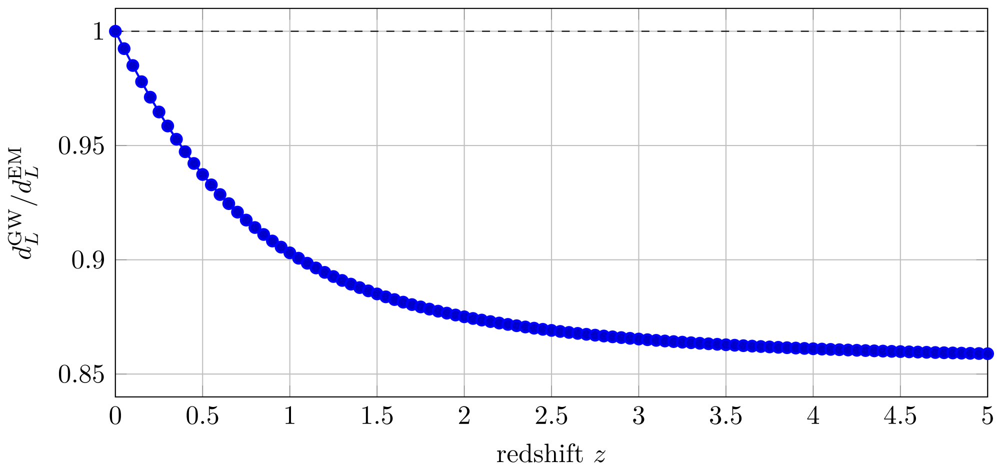

Figure 2: Standard-siren prediction evaluated from Eq. (25) on the homogeneous RFG background and rendered from numerical coordinates embedded in this LaTeX source, for $\Omega_{R0} = 0.70$ and $\theta = 1.6$. The ratio equals unity at $z = 0$, is approximately 0.903 at $z = 1$, and approaches $\sqrt{Q_T(0)} \approx 0.855$ at high redshift. No illustrative observational point is added, because a statistically meaningful comparison requires an explicit likelihood analysis.

12 RFG-R: regular multiscale effective completion

The original branch in Eqs. (17)–(18) is singular at $X = 0$ when $\theta > 0$. It therefore cannot by itself support a standard asymptotically flat weak-field expansion. This section defines a different, low-energy effective theory, called RFG-R. It is constructed to retain the cosmological branch to a controlled accuracy, to be regular at $X = 0$, and to state the local matching rule rather than silently extrapolating a homogeneous action into Solar-System geometry. RFG-R is a model definition; it is not claimed to follow uniquely from the original RFG action.

12.1 Regular cosmological action and exact reconstruction

Let p be an even integer satisfying $p > \theta$, let $\epsilon > 0$, and define

$$\nu := 1 + \frac{\theta}{p}, \qquad A := \frac{\Omega_{R0}}{1+\theta}. \tag{26}$$

The regular response and tensor normalization are

$$\mathcal{R}_\epsilon(X) := \Omega_{R0}\frac{X^{p+2}}{(X^p + \epsilon^p)^\nu}, \tag{27}$$

$$Q_\epsilon(X) := 1 - A\frac{X^p}{(X^p + \epsilon^p)^\nu}. \tag{28}$$

The cosmological RFG-R action is

$$S_{\cos} = \frac{M_{\rm Pl}^2}{2}\int \mathrm{d}^4x\sqrt{-g}\,\left[Q_\epsilon(X)\left({}^{(3)}R + K_{\mu\nu}K^{\mu\nu} - K^2\right) + 2H_0^2V_\epsilon(X)\right] + S_m[g,\Psi]. \tag{29}$$

Matter is minimally and universally coupled to the Jordan metric $g_{\mu\nu}$. Define

$$F_\epsilon(X) := X^2 - \mathcal{R}_\epsilon(X) - X^2Q_\epsilon(X) - X^3Q_{\epsilon,X}(X). \tag{30}$$

The FLRW lapse equation has the target form

$$\boxed{E^2 = \Omega_{m0}a^{-3} + \Omega_{r0}a^{-4} + \mathcal{R}_\epsilon(E).} \tag{31}$$

Comparison with Eq. (20) fixes the potential, up to the usual ADM boundary representative CX, through

$$V_\epsilon - XV_{\epsilon,X} = 3F_\epsilon(X), \tag{32}$$

$$\boxed{V_\epsilon(X) = -3X\int_0^X \frac{F_\epsilon(s)}{s^2}\,\mathrm{d}s.} \tag{33}$$

The integrand is finite at the origin. Thus Eq. (33) is an action-level reconstruction, not a background-only prescription.

For $X \gg \epsilon$,

$$\mathcal{R}_\epsilon(X) = \Omega_{R0}X^{2-\theta}\left[1 - \left(1+\frac{\theta}{p}\right)\left(\frac{\epsilon}{X}\right)^p + O\left((\epsilon/X)^{2p}\right)\right], \tag{34}$$

$$Q_\epsilon(X) = 1 - AX^{-\theta}\left[1 - \left(1+\frac{\theta}{p}\right)\left(\frac{\epsilon}{X}\right)^p + O\left((\epsilon/X)^{2p}\right)\right]. \tag{35}$$

The reference choice $p = 4$, $\epsilon = 10^{-8}$, and $\theta = 1.6$ therefore changes the original branch by at most 2.5×10^{-32} for $X \geq 0.8$, analytically. At small X, $Q_\epsilon(0) = 1$ and $V_\epsilon = O(X^{p+2})$, so the Einstein–Hilbert ADM kinetic operator is recovered at $K = 0$. Figure 3 displays the numerical recovery and regularization check. The resulting radiation, matter, and relational fractions on the same regularized expanding branch are shown in Fig. 4.

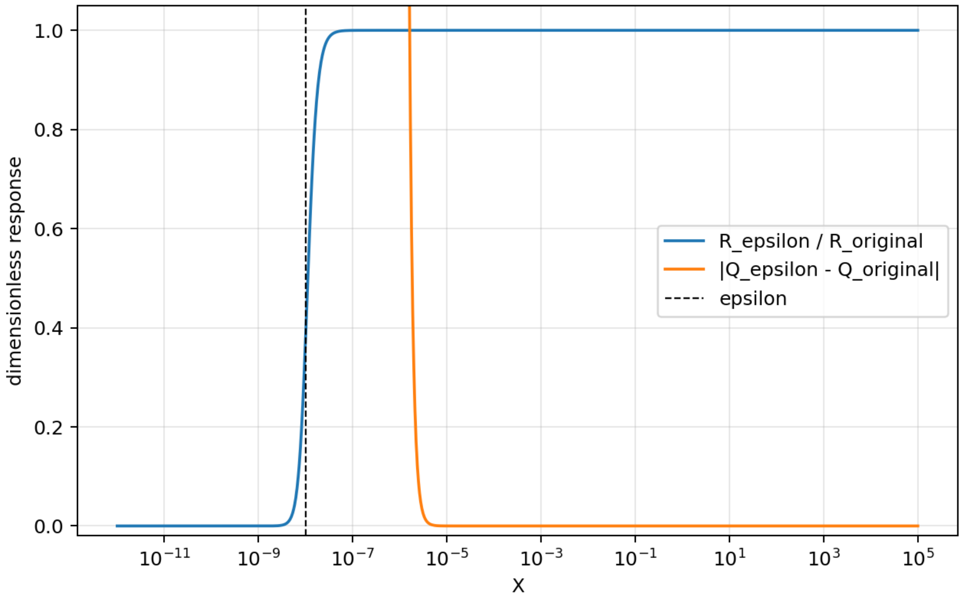

Figure 3: RFG-R regularization check. The response ratio $\mathcal{R}_\epsilon/\mathcal{R}_{\text{original}}$ and the tensor-normalization difference $|Q_\epsilon - Q_{\text{original}}|$ are evaluated with $p = 4$, $\epsilon = 10^{-8}$, and $\theta = 1.6$. Above the regulated $X \simeq \epsilon$ domain, RFG-R recovers the original cosmological branch; the analytic high-X error is given in Eq. (35). This is an action-level numerical validation, not an observational fit. The generating script is `output generator`.

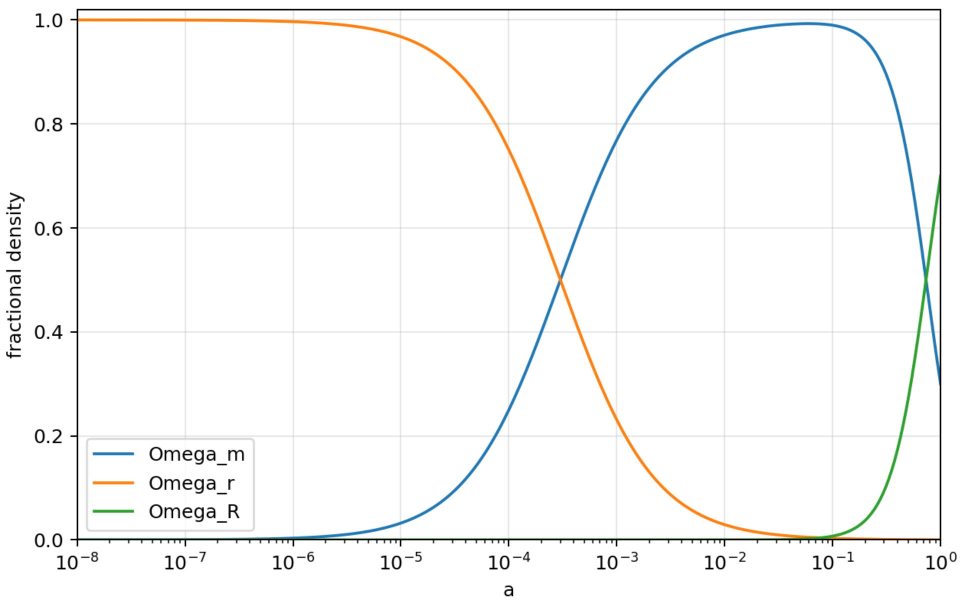

Figure 4: Fractional densities on the regularized RFG-R expanding branch of Eq. (31), with $\Omega_{m0} = 0.30$, $\Omega_{r0} = 9 \times 10^{-5}$, $\Omega_{R0} = 0.69991$, $\theta = 1.6$, $p = 4$, and $\epsilon = 10^{-8}$. The curves show radiation, matter, and relational components computed from the action-reconstructed background equation. They are not a compressed CMB constraint. The generating script is `output generator`.

12.2 Local GR matching and PPN statement

The cosmological action is supplemented by an explicit multiscale matching rule. Let

$$W := \frac{\sqrt{|C_{\alpha\beta\gamma\delta}C^{\alpha\beta\gamma\delta}|}}{(H_0/c)^2}. \tag{36}$$

Choose a C^∞ switch $s(W)$ with $s = 1$ for $W \leq 10^8$ and $s = 0$ for $W \geq 10^9$; its explicit bump-function construction is given in `completion derivation`. The complete low-energy EFT is

$$S_{\text{eff}} = S_{\text{GR}} + \int \mathrm{d}^4x \, s(W) \, (\mathcal{L}_{\cos} - \mathcal{L}_{\text{GR}}) \, . \tag{37}$$

Here $\mathcal{L}$ denotes the corresponding covariant action density, including $\sqrt{-g}$. Flat FLRW has $W = 0$, hence uses the cosmological action. In every open region with $W \geq 10^9$, both s and all its derivatives vanish, so the action and all its variations are exactly those of GR with the same universally coupled matter sector. For a Schwarzschild Sun,

$$W(r) = \frac{\sqrt{48}\, GM_\odot/c^2}{r^3(H_0/c)^2}, \qquad W(1\,\text{AU}) = 5.756 \times 10^{22} \tag{38}$$

at the stated reference point. The model prediction in that local domain is therefore

$$\boxed{\gamma_{\text{PPN}} = \beta_{\text{PPN}} = 1, \qquad \alpha_1 = \alpha_2 = 0.} \tag{39}$$

This is an EFT matching prediction, not a derivation of a microscopic screening mechanism. The packaged Cassini Gaussian factor gives $-2\ln\mathcal{L}_{\text{Cassini}} = 0.833648$ for Eq. (39) and the reported measurement [22].

12.3 Tensor and matter/CMB likelihood interface

On FLRW, the tensor result is obtained from the same quadratic reduction as before with Q replaced by Q_ϵ:

$$Q_T(a) = Q_\epsilon(E(a)) > 0, \qquad c_T^2 = 1, \qquad \frac{d_L^{\text{GW}}}{d_L^{\text{EM}}} = \sqrt{\frac{Q_T(0)}{Q_T(z)}}. \tag{40}$$

The reference validation grid has $\min Q_T = 0.6153848$. The time evolution of Q_T and of the corresponding Planck-mass running α_M is shown in Fig. 5; the positive tensor normalization is the no-ghost check behind Eq. (40).

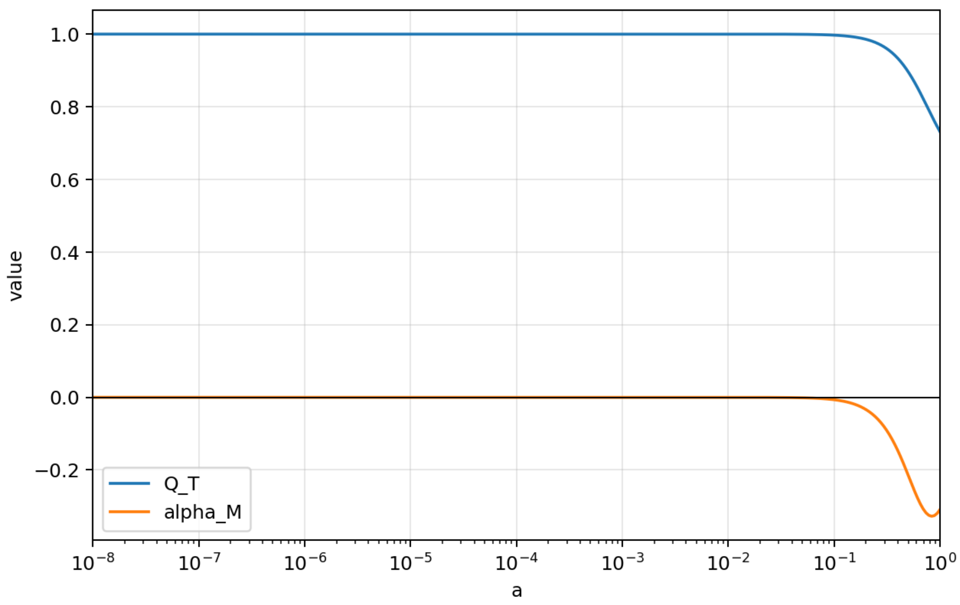

Figure 5: RFG-R tensor kinetic normalization Q_T and Planck-mass running α_M as functions of scale factor on the reference branch. The positive Q_T curve verifies the tensor no-ghost condition, while the quadratic tensor action gives $c_T = 1$ exactly. The plot defines the standard-siren propagation prediction in Eq. (40); it is not itself a gravitational-wave or CMB likelihood evaluation. The generating script is `output generator`.

The matter and CMB test is defined from the action derivatives, not from an assumed quasi-static $\mu(a,k)$. In particular,

$$\mathcal{L}_{^{(3)}R} = Q_\epsilon, \tag{41}$$

$$\mathcal{L}_{K_{ij}{}^{(3)}R} = \frac{Q_{\epsilon,X}}{3H_0}\gamma^{ij}, \tag{42}$$

$$\mathcal{L}_{K_{ij}} = 2Q_\epsilon(K^{ij} - K\gamma^{ij}) + \left[\frac{Q_{\epsilon,X}}{3H_0}\left({}^{(3)}R + K_{mn}K^{mn} - K^2\right) + \frac{2H_0}{3}V_{\epsilon,X}\right]\gamma^{ij}. \tag{43}$$

12.3.1 Exact ADM-to-extended-EFT map and solver boundary

The preceding derivatives can be translated into a perturbation basis without fitting a background surrogate. The extended-EFT map of Ref. [12] uses the opposite extrinsic-curvature sign from the convention of this manuscript. Thus, for the map only,

$$K_{ij}^{\rm map} = -K_{ij}^{\rm RFG}, \qquad X = -\frac{K^{\rm map}}{3H_0}. \tag{44}$$

With this explicit conversion, the exact RFG-R action gives

$$\begin{aligned} 1 + \Omega_{\rm EFT} &= Q_\epsilon, & \bar{M}_1^3 &= -M_{\rm Pl}^2\dot{Q}_\epsilon, \\ \frac{\bar{M}_2^2}{M_{\rm Pl}^2} &= \frac{X^2 Q_{\epsilon,XX}}{3} + \frac{4XQ_{\epsilon,X}}{3} - \frac{V_{\epsilon,XX}}{9}, & \bar{M}_3^2 &= 0, \\ \bar{m}_5 &= -\frac{M_{\rm Pl}^2 Q_{\epsilon,X}}{3H_0}. \end{aligned} \tag{45}$$

In particular, the last coefficient multiplies

$$\frac{\bar{m}_5}{2}\,\delta^{(3)}R\,\delta K = -\frac{M_{\rm Pl}^2 Q_{\epsilon,X}}{6H_0}\,\delta^{(3)}R\,\delta K. \tag{46}$$

It is required by the $Q_\epsilon(K)^{(3)}R$ term in the action, hence cannot consistently be set to zero. The executable `mapping note`, `generator`, and `CSV table` are public and mutually linked. On the published mapping grid $10^{-7} \le a \le 1$, the independent validation gives $\min Q_\epsilon = 0.7308038462$, $\max|H_0\bar{m}_5/M_{\rm Pl}^2| = 0.1435712821$, a maximum finite-difference error 8.018×10^{-7} for the second implicit background derivative, and zero numerical error in the algebraic $\bar{m}_5$ identity.

12.3.2 Executed pure-gravity extended-EFT scalar audit

The scalar action must also be checked in the same operator basis before it is handed to a Boltzmann hierarchy. For the exact RFG-R map, $m_2^2 = \lambda_i = \hat{M}^2 = 0$, while both $\bar{m}_5$ and $W_7 = -(\bar{M}_3^2 + \bar{M}_2^2)/(2a^4)$ are nonzero. Eliminating the lapse and shift in the pure-gravity unitary-gauge action of Ref. [12] gives

$$L_{\dot\zeta\dot\zeta} = \frac{(6a^2W_7 + W_5)(3a^4W_4^2 + 2a^2W_1W_5)}{2a^2(W_4^2 - 4W_1W_7)}. \tag{47}$$

The full $\bar{m}_5$ dependence is retained in the gradient and mixing terms; its absence from Eq. (47) is not a licence to delete the operator. The executable `scalar audit`, its independent `regression test`, the complete `derivation note`, and the machine-readable `audit table` are public. On a 49×49 logarithmic grid $10^{-7} \le a \le 1$ and $10^{-4} \le k/H_0 \le 10^5$, all 2401 points satisfy

$$\mathcal{A}_K := 6a^2W_7 + W_5, \qquad \mathcal{B}_+ := 3a^4W_4^2, \qquad \mathcal{B}_- := 2a^2W_1W_5, \tag{48}$$

$$\min(W_4^2 - 4W_1W_7) = 2.514017848646 > 0, \qquad \max \frac{|\mathcal{A}_K(\mathcal{B}_+ + \mathcal{B}_-)|}{|\mathcal{A}_K|\,(|\mathcal{B}_+| + |\mathcal{B}_-|)} = 4.357 \times 10^{-16}. \tag{49}$$

Thus the pure-gravity scalar quadratic kinetic coefficient is degenerate to the precision of the calculation; a standalone scalar sound speed is undefined. This is a pre-Boltzmann strong-coupling/constraint gate, not a successful scalar-stability result. It does *not* by itself prove that the physical matter-coupled theory is ill posed: matter sources the lapse and shift constraints, and the separate one-canonical-scalar reduction below has a different, nondegenerate sourced Hessian. The decisive missing object is therefore the complete photon–baryon–CDM–neutrino kinetic and gradient matrix derived from the same action.

The audited public H-EFTCAMB source exposes the usual $\Omega, \gamma_1, \ldots, \gamma_6$ interface, but not the coefficient in Eq. (46); the source-level audit is public in `the backend audit`. It therefore cannot produce an exact RFG-R CMB, lensing, or matter spectrum without an extension of its scalar equations and stability checks. Independently of that backend gap, Eq. (49) blocks a direct one-scalar EFT run. This is a precise implementation boundary, not a negative data result and not a licence to omit the operator. The full kinetic Hessian must be obtained by differentiating Eq. (43) after the physical matter species are included. For every sampled cosmology, a full 3+1 Einstein–Boltzmann implementation must reject absent background roots, $Q_T \le 0$, ghosts, gradient instabilities, and singular constraint matrices before computing $C_\ell^{TT,TE,EE}$, $C_L^{\phi\phi}$, $P(k,z)$, $f\sigma_8(z)$, and Eq. (40). The intended joint likelihood is

$$\ln\mathcal{L} = \ln\mathcal{L}_{\rm Planck18} + \ln\mathcal{L}_{\rm lensing} + \ln\mathcal{L}_{\rm BAO} + \ln\mathcal{L}_{\rm SN} + \ln\mathcal{L}_{\rm RSD} + \ln\mathcal{L}_{\rm PPN} + \ln\mathcal{L}_{\rm GW}. \tag{50}$$

The public calculation protocol, exact input/output contract, and deterministic rejection conditions are supplied in `likelihood protocol`. The installed Planck likelihood components provide the data interface [16]; an RFG-R solver must implement the physical multi-fluid reduction, retain Eq. (46), reproduce its GR limit, and then be coupled to that interface. This is intentionally not a claim that the likelihood in Eq. (50) has already been evaluated.

12.3.3 Executed GR data-interface reference

The ordinary radiation, recombination, lensing and data-interface machinery has been exercised separately in GR using CAMB [11] version 2.0.1 and the Planck 2018 TT,TE,EE+lowE+lensing posterior-mean cosmological inputs $H_0 = 67.36\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, $\omega_b = 0.02237$, $\omega_c = 0.1200$, $\tau = 0.0544$, $\ln(10^{10}A_s) = 3.044$, $n_s = 0.9649$, and a BBN-consistent helium fraction [9]. The pinned numerical reference gives $\sigma_8(0) = 0.8110325278646$ together with reproducible lensed TT, TE, EE, lensing-potential, and linear-matter samples in `the GR CAMB CSV`. The exact generator and regression test are `the CAMB generator` and `its regression test`.

The official Planck low-temperature, low-polarization and lensing components were then evaluated at that fixed GR point. Their directly computed values are

$$\ln \mathcal{L}^{\mathrm{GR}}_{\mathrm{lowT}} = -11.7117375873, \qquad \ln \mathcal{L}^{\mathrm{GR}}_{\mathrm{lowE}} = -198.063447867, \qquad \ln \mathcal{L}^{\mathrm{GR}}_{\mathrm{lens}} = -4.39556885539, \tag{51}$$

or $-2\ln \mathcal{L}^{\mathrm{GR}}_{\mathrm{low}\ell+\mathrm{lens}} = 428.341508619$. The machine result, execution script and regression test are public in `the likelihood CSV`, `the likelihood script`, and `its regression test`. The high-ℓ Plik package is installed but deliberately not evaluated with guessed foreground and calibration parameters: those form part of a documented joint fit. This reference validates the public data path only. It sets $\bar{m}_5 = 0$ because CAMB is GR, and is therefore neither an RFG-R likelihood nor evidence for RFG-R against ΛCDM; see `the data-reference note`.

12.3.4 Exact photon–baryon–CDM–neutrino reduction

The physical sourced reduction is obtained directly from the RFG-R action, rather than by adding a phenomenological $\mu(a,k)$, a sound-speed closure, or a quasi-static approximation. For each barotropic monopole–dipole matter source, the starting point is the Sorkin–Schutz action [13]

$$S_i = -\int \mathrm{d}^4x \left[\sqrt{-g}\,\rho_i(n_i) + J_i^\mu \partial_\mu \ell_i\right], \qquad p_i = n_i\rho_{i,n} - \rho_i. \tag{52}$$

Let $s = k^2\psi/a^2$, and define the physical and auxiliary vectors

$$\mathbf{q} = (\zeta, \delta_b, \delta_c, \delta_\gamma, \delta_\nu)^T, \qquad \mathbf{x} = (\alpha, s, v_b, v_c, v_\gamma, v_\nu)^T. \tag{53}$$

The exact finite scalar density, retaining the complete extended-EFT gravity action, is

$$L^{(2)} = \frac{1}{2}\dot{\mathbf{q}}^T K_0 \dot{\mathbf{q}} + \mathbf{x}^T C \dot{\mathbf{q}} + \frac{1}{2}\mathbf{x}^T A \mathbf{x} + \mathbf{x}^T D \mathbf{q} + \frac{1}{2}\mathbf{q}^T M_0 \mathbf{q}. \tag{54}$$

Its nonzero entries are

$$\begin{aligned}
(K_0)_{\zeta\zeta} &= -3a^2W_5, & C_{\alpha\zeta} &= -3a^2W_4, & C_{s\zeta} &= -a^2W_5,\\
A_{\alpha\alpha} &= 2W_1, & A_{\alpha s} &= -a^2W_4, & A_{ss} &= 2a^4W_7,\\
D_{\alpha\zeta} &= -W_6k^2, & D_{s\zeta} &= -2\bar{m}_5\frac{k^2}{a^2}, & (M_0)_{\zeta\zeta} &= -2W_0k^2,\\
C_{v_i\zeta} &= -3\rho_i(1+w_i), & C_{v_i\delta_i} &= -\rho_i, & A_{sv_i} &= -\rho_i(1+w_i),\\
A_{v_iv_i} &= -\rho_i(1+w_i)\frac{k^2}{a^2}, & D_{\alpha\delta_i} &= -\rho_i, & (M_0)_{\delta_i\delta_i} &= -\frac{\rho_i w_i}{1+w_i}.
\end{aligned} \tag{55}$$

Here $i = b, c, \gamma, \nu$, with $w_{b,c} = 0$ and $w_{\gamma,\nu} = 1/3$ only in the density–momentum constraint rows. It is not a closure of the photon or neutrino hierarchy.

Varying α and s gives the exact sourced lapse–shift system

$$
\begin{aligned}
2W_1\alpha - a^2W_4 s - 3a^2W_4\dot{\zeta} - W_6k^2\zeta - \sum_i \rho_i\delta_i = 0,\\
-a^2W_4\alpha + 2a^4W_7 s - a^2W_5\dot{\zeta} - 2\bar{m}_5\frac{k^2}{a^2}\zeta - \sum_i \rho_i(1+w_i)v_i = 0.
\end{aligned}
\tag{56}
$$

The determinant is the analytic identity

$$
\det A_{(\alpha,s)} = -a^4\left(W_4^2 - 4W_1W_7\right), \tag{57}
$$

and eliminating all auxiliary variables yields the complete finite Schur reduction

$$
K = K_0 - C^TA^{-1}C, \qquad B = -C^TA^{-1}D, \qquad M = M_0 - D^TA^{-1}D. \tag{58}
$$

The photon and neutrino anisotropic-stress sectors are not discarded. In the same spatial gauge, $\chi = -\psi$, their exact massless Boltzmann hierarchy is [14]

$$
\dot{\Theta}_\ell = \frac{k}{a}\left[\frac{\Theta_{\ell-1}}{2\ell-1} - \frac{\Theta_{\ell+1}}{2\ell+3}\right] + M_\ell + C_\ell, \tag{59}
$$

where the $\ell = 0$ streaming term is $-k\Theta_1/(3a)$, and

$$
M_0 = -\dot{\zeta} - \frac{s}{3}, \qquad M_1 = \frac{k}{a}\alpha, \qquad M_2 = \frac{2s}{3}, \qquad M_{\ell\geq 3} = 0. \tag{60}
$$

For photons $C_0 = 0$, $C_1 = \dot{\tau}(v_b - \Theta_1)$, $C_2 = \dot{\tau}(P - \Theta_2)$, $C_{\ell\geq3} = -\dot{\tau}\Theta_\ell$, where

$$
P = \frac{\Theta_2 - \sqrt{6}E_2}{10}, \qquad \dot{\tau} = n_e x_e \sigma_T \geq 0. \tag{61}
$$

The exact scalar polarization hierarchy, for $\ell \geq 2$, is

$$
\dot{E}_\ell = \frac{k}{a}\left[\frac{\sqrt{\ell^2-4}}{2\ell-1}E_{\ell-1} - \frac{\sqrt{(\ell+1)^2-4}}{2\ell+3}E_{\ell+1}\right] - \dot{\tau}\left(E_\ell + \sqrt{6}P\,\delta_{\ell 2}\right), \qquad B_\ell^{(0)} = 0. \tag{62}
$$

For a collisionless massless neutrino, $C_\ell = 0$ at every ℓ. Thus Eqs. (59)–(62) are an infinite hierarchy, not a fluid approximation or a prescribed $\ell_{\max}$ terminal closure. Their first moments obey $\delta_{\gamma,\nu} = 4\Theta_{\gamma,\nu,0}$, $v_{\gamma,\nu} = \Theta_{\gamma,\nu,1}$, and $\pi_{\gamma,\nu}/\rho_{\gamma,\nu} = 4\Theta_{\gamma,\nu,2}/5$, which matches the constraint action above exactly.

For a massive neutrino no massless substitution is made. With $\epsilon(q,a) = \sqrt{q^2 + a^2m_\nu^2}$, the collisionless phase-space equation is

$$
\delta f' = -\frac{q}{\epsilon}\mathcal{D}_\gamma\delta f + q\frac{\partial f_0}{\partial q}\left[\frac{\epsilon}{q}A_{,\alpha}\gamma^\alpha + (B_{\alpha|\beta} + C'_{\alpha\beta})\gamma^\alpha\gamma^\beta\right]. \tag{63}
$$

The density, momentum and anisotropic stress are the corresponding exact momentum integrals with the ϵ weights; no effective equation of state is inserted.

The executable derivation is `the full reduction note`; the matrix/hierarchy implementation is `the reduction module`, and the independent validation is `the audit script`. The GR test is performed with rational arithmetic and gives rank $K = 4$ exactly. The separate 425-point RFG-R audit uses the published Planck 2018 Table 2 species split solely as a documented input record [9]: it gives a maximum constraint-determinant residual 4.148×10^{-15} and inertia $(n_+, n_-, n_0) = (4, 0, 1)$ at every point. Those values are not an RFG-R fit, and neither this reduction nor the audit evaluates a CMB likelihood. The machine table is `multifluid audit CSV`.

12.3.5 Spatial metric equations and exact DAE closure

The lapse–shift reduction alone is not a metric evolution law. To obtain the remaining scalar equation, the traceless spatial deformation must be retained before the spatial gauge is imposed:

$$\gamma_{ij} = a^2 e^{2\zeta} \exp(2D_{ij}E), \qquad D_{ij} = \partial_i\partial_j - \frac{1}{3}\delta_{ij}\partial^2. \tag{64}$$

Direct variation of the original ADM action with respect to E, followed only then by $E = 0$, gives in units $H_0 = M_{\rm Pl} = 1$, with $s = k^2\beta/a^2$,

$$\dot{s} = -\left(3H + \frac{\dot{Q}}{Q} + \frac{Q_X k^2}{3Qa^2}\right) s - \frac{k^2}{a^2}(\alpha+\zeta) + \frac{Q_X k^2}{Qa^2}(H\alpha - \dot{\zeta}) - \frac{\Pi}{Q}. \tag{65}$$

Here Π is the total scalar anisotropic stress; for each massless kinetic species $\Pi_i = 4\rho_i\Theta_{i2}/5$. The independently checked Einstein–Hilbert limit of Eq. (65) is

$$\dot{s} = -3Hs - \frac{k^2}{a^2}(\alpha+\zeta) - \Pi. \tag{66}$$

The executable variation performs the lapse, shift, trace, and traceless GR regressions directly from the ADM density: `spatial-variation script`.

The exact Schur reduction in Eq. (58) has one null kinetic direction. It becomes explicit under the invertible field redefinition

$$\Delta_i = \delta_i + 3(1+w_i)\zeta, \qquad \mathbf{y} = (\zeta, \Delta_b, \Delta_c, \Delta_\gamma, \Delta_\nu)^T. \tag{67}$$

Writing the reduced density as

$$L_{\rm red} = \frac{1}{2}\dot{\mathbf{y}}^T K_\Delta \dot{\mathbf{y}} + \dot{\mathbf{y}}^T B_\Delta \mathbf{y} + \frac{1}{2}\mathbf{y}^T M_\Delta \mathbf{y}, \tag{68}$$

the exact identities are

$$(K_\Delta)_{0j} = 0, \qquad B_\Delta = B_\Delta^T. \tag{69}$$

Thus ζ is not an omitted propagating mode and no temporal gauge has been selected: its Euler equation is the algebraic curvature constraint

$$\mu_\zeta(a,k)\zeta + \sum_i \left[(M_\Delta)_{0i} - (\dot{B}_\Delta)_{0i}\right]\Delta_i = 0, \qquad \mu_\zeta = (M_\Delta)_{00} - (\dot{B}_\Delta)_{00}. \tag{70}$$

All derivatives in Eq. (70) are analytic derivatives of the RFG-R action, including Q_{XXX}, V_{XXX}, and $\dot{W}_i$; no tabulated-background finite difference enters the result.

This calculation closes the finite DAE question, and it gives a negative viability result for the stated reference branch. On a 49-point logarithmic grid in $a \in [10^{-7}, 1]$, 24 scale factors have a finite positive root of μ_ζ. In particular,

$$\left.\frac{k}{H_0}\right|_{a=1,\ \mu_\zeta=0} = 2.51545672221. \tag{71}$$

At this surface Eq. (70) ceases to determine ζ. The independently validated residuals are $\max_j |(K_\Delta)_{0j}|/\max|K_\Delta| = 5.923\times10^{-15}$ and $\max|B_\Delta - B_\Delta^T|/\max|B_\Delta| = 1.360\times10^{-16}$, so the zero is not a hierarchy truncation or a loss of numerical precision. The derivation, independent regression, and root table are public in `the DAE derivation`, `the DAE generator`, `its independent validator`, and `the root table`.

Therefore the published RFG-R reference action does not define globally regular linear transfer functions. No CMB, lensing, matter, or joint likelihood value for that branch may be obtained by forcing an integration through Eq. (71). A modified action would be a new theory and would require a fresh derivation and validation.

12.3.6 RFG-RΞ: background-preserving foliation completion

The singularity in Eq. (70) is an internal property of the reference action, not a request to revert to ΛCDM. The minimal background-preserving R-Universe extension considered here is a new action,

$$S_\Xi = S_{\rm cos} + \frac{M_{\rm Pl}^2}{2}\int {\rm d}^4x\, N\sqrt{\gamma}\,\Xi\left({}^{(3)}R + \sigma_{ij}\sigma^{ij}\right), \qquad \Xi = \text{constant}, \tag{72}$$

where $\sigma_{ij} = K_{ij} - K\gamma_{ij}/3$. Both terms in Eq. (72) vanish on a spatially flat FLRW background. The reconstructed R-Universe Friedmann equation (31) is therefore exactly unchanged. The same term adds equally to the tensor kinetic and gradient coefficients,

$$Q_T^\Xi = Q_\epsilon + \Xi, \qquad c_T^2 = 1. \tag{73}$$

The existing smooth Weyl matching rule multiplies the entire cosmological Lagrangian, including Eq. (72); it remains exactly GR in the local constant-switch domain.

For a real scalar Fourier mode, direct expansion of the ADM action gives

$$\frac{\Delta L_\Xi^{(2)}}{a^3} = \Xi\left[2\frac{k^2}{a^2}\alpha\zeta + \frac{k^2}{a^2}\zeta^2 + \frac{s^2}{3}\right]. \tag{74}$$

Thus the finite matrices of Eq. (54) are modified without a fitted background function:

$$\Delta A_{ss} = \frac{2\Xi}{3}, \qquad \Delta D_{\alpha\zeta} = \Delta(M_0)_{\zeta\zeta} = 2\Xi\frac{k^2}{a^2}, \qquad \partial_t\Delta D_{\alpha\zeta} = \partial_t\Delta(M_0)_{\zeta\zeta} = -2H\Delta D_{\alpha\zeta}. \tag{75}$$

Equations (72)–(75) define RFG-RΞ, not a reinterpretation of the singular reference action. Keeping the scalar traceless metric mode until after variation gives the following exact additions to the lapse, shift, trace, and traceless Euler residuals:

$$\begin{aligned} \Delta\mathcal{E}_\alpha &= 2\Xi a k^2\zeta, & \Delta\mathcal{E}_\beta &= \frac{2\Xi\beta k^4}{3a}, \\ \Delta\mathcal{E}_\zeta &= -2\Xi a k^2(\alpha+\zeta), & \Delta\mathcal{E}_E &= -\frac{2\Xi a k^4}{3}\left(H\beta + \alpha + \dot\beta + \zeta\right). \end{aligned} \tag{76}$$

Thus this is not an Einstein-equation substitution. In the gauge $s = k^2\beta/a^2$, the directly varied traceless equation is

$$\begin{aligned} \dot s = &-\left[3H + \frac{\dot Q_\epsilon}{Q_\epsilon+\Xi} + \frac{Q_{\epsilon,X}}{3(Q_\epsilon+\Xi)}\frac{k^2}{a^2}\right]s - \frac{k^2}{a^2}(\alpha+\zeta) \\ &+ \frac{Q_{\epsilon,X}}{Q_\epsilon+\Xi}\frac{k^2}{a^2}\left(H\alpha - \dot\zeta\right) - \frac{\Pi}{Q_\epsilon+\Xi}. \end{aligned} \tag{77}$$

Here Π is the total kinetic anisotropic stress. The symbolic regression of all four residuals and the executable form of Eq. (77) are in `the RFG-RXi metric-equation test`.

The constant Ξ is a new dimensionless EFT coefficient, not a fitted cosmological parameter in this work. For the unfitted normalization benchmark $\Xi = 1$, the exact multi-fluid DAE audit scans 49 values of $a \in [10^{-7}, 1]$ and 81 values of $k/H_0 \in [10^{-4}, 10^6]$. It finds no sampled μ_ζ root, $\min Q_T^\Xi = 1.736101654462$, and $\min|\det A_{(\alpha,s)}| = 5.991445560907$. The four-dimensional material kinetic block has inertia $(4,0,0)$ at every audited point. The direct symbolic identity, audit, and table are reproducible from `the RFG-RXi derivation`, `the RFG-RXi audit`, `its regression test`, `the exact metric-equation regression`, and `the audit table`. An extended audit samples 193 values of $a \in [10^{-8}, 1]$ and 601 values of $k/H_0 \in [10^{-5}, 10^7]$ for each of $\Xi = 1, 2$. It finds no sampled curvature-constraint root, retains positive Q_T^Ξ and a nonsingular lapse–shift block, and has $\min|\mu_\zeta|_{\rm normalized} = 1$ on that declared domain. Its generator, independent check, and machine table are `the extended DAE audit`, `its regression test`, and `the extended audit table`.

Directly evaluated RFG-RΞ factors. The local Weyl switch is exactly zero in the Solar-System matching domain, so the Ξ operator is absent there and the action gives

$$\gamma_{\rm PPN} = \beta_{\rm PPN} = 1, \qquad \alpha_1 = \alpha_2 = 0, \qquad -2\ln\mathcal{L}_{\rm Cassini} = 0.8336483932. \tag{78}$$

This is a directly evaluated Cassini data factor for both $\Xi = 1$ and $\Xi = 2$, not a surrogate for a CMB likelihood. The corresponding exact tensor prediction is

$$Q_T^\Xi(a) = Q_\epsilon(a) + \Xi, \qquad c_T^2 = 1, \qquad \frac{d_L^{\rm GW}}{d_L^{\rm EM}} = \left[\frac{Q_T^\Xi(1)}{Q_T^\Xi(a)}\right]^{1/2}. \tag{79}$$

At the Planck-recorded background input, the action yields $d_L^{\rm GW}/d_L^{\rm EM} = 0.9565968846$ at $z \simeq 1$ for $\Xi = 1$, and 0.9717969466 for $\Xi = 2$. The full grid, the exact Cassini factor, and the linked DAE diagnostics are public in `the direct-factor note`, `the observable generator`, `its regression test`, `the observable grid`, and `the factor summary`.

This is a viability gate for a distinct R-Universe action, not a CMB fit, posterior, or a proof of nonlinear health. Initial conditions, the complete Boltzmann integration, gradient and nonlinear stability, and an observational likelihood remain to be calculated from Eq. (72).

12.4 Exact canonical-scalar reference reduction

The first sourced scalar check for the regular RFG-R action can be completed without a quasi-static approximation. It is deliberately narrower than a cosmological matter calculation: the matter sector is one minimally coupled, canonical scalar ϕ, and scalar-field comoving gauge $\delta\phi = 0$ is used. Thus the result tests the physical scalar mode and the sourced lapse–shift constraints, but it is not a replacement for the photon, baryon, cold-dark-matter and neutrino hierarchies required by the CMB likelihood.

Put $p = k^2/a^2$, $y = p\beta$, $m = \dot\phi^2/M_{\rm Pl}^2$, and $D = -F_{\epsilon,XX}/3$, where F_ϵ is the RFG-R ADM Lagrangian evaluated on the background. After including the complete intrinsic-curvature contribution and the canonical scalar action, the nondynamical variables $\mathbf{x} = (\alpha, y)^T$ have Hessian

$$H_{(\alpha,y)} = \begin{pmatrix} 2(-3DH^2 + m) & 2DH \\ 2DH & 2(-D + 2Q_\epsilon)/3 \end{pmatrix}, \qquad \det H_{(\alpha,y)} = -\frac{4}{3}\Delta, \tag{80}$$

with the exact constraint discriminant

$$\Delta = 6DH^2Q_\epsilon + (D - 2Q_\epsilon)m. \tag{81}$$

The lapse and shift constraints are invertible precisely where $\Delta \neq 0$. Eliminating them by the Schur complement gives

$$S_{\rm red}^{(2)} = \frac{M_{\rm Pl}^2}{2}\int {\rm d}t\,{\rm d}^3k\, a^3 \left[K\dot\zeta^2 + \left(C_0 + C_1 p + C_2 p^2\right)\zeta^2\right], \qquad K = \frac{6DQ_\epsilon m}{6DH^2Q_\epsilon + (D - 2Q_\epsilon)m}. \tag{82}$$

Here the mixed $\zeta\dot\zeta$ term has been integrated by parts; the exact algebraic expressions for C_0, C_1, C_2 follow from the same matrix reduction. The physical low-gradient and k^4 coefficients are

$$G_1 = -C_1, \qquad G_2 = -C_2, \qquad c_{s,\rm low}^2 = \frac{G_1}{K} \quad (K > 0). \tag{83}$$

Equations (80)–(83) are exact for the stated one-field system. The full derivation, including the curvature term needed to recover the GR sound speed, is available in `canonical-scalar derivation`; the executable calculation is `canonical-scalar solver` and its machine readable reference output is `reference CSV`.

Two checks are built into the calculation. First, GR coupled to a massless canonical (stiff) scalar gives exactly $K = 6$ and $c^2_{s,\mathrm{low}} = 1$. Second, on the regular RFG-R reference branch $\theta = 1.6$, $\Omega_{\phi 0} = 0.30$, $\Omega_{R0} = 0.70$, and $0.03 \le a \le 1$, the numerical scan returns

$$\begin{aligned} \min \Delta &= 8.006769231, & \min K &= 1.695411575, \\ \min G_1 &= 0.5673484131, & c^2_{s,\mathrm{low}}(a=1) &= 0.3346375721. \end{aligned} \tag{84}$$

The calculated G_2 is non-negative to numerical precision; its minimum is -4.7×10^{-10}, a floating-point residual around the GR-asymptotic value zero. This is a reproducible internal viability gate for the specified one-field branch, not a multi-fluid stability theorem or a CMB data result.

12.5 Reproducibility and public links

The self-contained supplement distributed beside this LaTeX source is `README`. Its calculation manifest is `calculation manifest`; the executable validation command is `bash r_universe_completion/scripts/run_all.sh`. The permanent public counterparts are linked directly from the paper: `completion derivation`, `likelihood protocol`, `multifluid reduction`, and `canonical-scalar derivation`, `multifluid module`, `multifluid audit`, `canonical-scalar solver`, and `validation script`. The repository directory also contains the exact source of the manuscript, generated tables, figures, and the `standalone RFG-R completion article`.

13 Quadratic scalar geometry and constraint kernel

This section develops the scalar perturbation sector directly from the RFG action. No auxiliary scalar field and no scalar–tensor closure assumption is introduced. The only inputs are the preferred-foliation ADM variables and the functions $Q(X)$ and $V(X)$ already fixed by the homogeneous reconstruction.

We use

$$N = 1 + \alpha, \qquad N_i = \partial_i \beta, \qquad \gamma_{ij} = a^2 e^{2\zeta} \delta_{ij}, \tag{85}$$

and

$$K_{ij} = \frac{1}{2N}\left(\dot\gamma_{ij} - D_i N_j - D_j N_i\right), \qquad X = \frac{K}{3H_0}. \tag{86}$$

All perturbative orders below refer to powers of α, β, ζ.

13.1 Exact measure and intrinsic curvature

Lemma 13.1 (Measure expansion)**.** *For the parametrization* (85)*,*

$$\sqrt{\gamma} = a^3 e^{3\zeta}, \tag{87}$$

and hence

$$N\sqrt{\gamma} = a^3\left[1 + \alpha + 3\zeta + 3\alpha\zeta + \frac{9}{2}\zeta^2\right] + O(3). \tag{88}$$

Proof. The determinant is $\det \gamma_{ij} = a^6 e^{6\zeta}$. Taking its positive square root and expanding the exponential gives (88). Because $N = 1 + \alpha$ is used as the perturbation variable, no independent α^2 term occurs in the measure. □

Lemma 13.2 (Intrinsic curvature expansion)**.** *For* $\gamma_{ij} = a^2 e^{2\zeta}\delta_{ij}$*,*

$$^{(3)}R = \frac{e^{-2\zeta}}{a^2}\left[-4\partial^2\zeta - 2(\partial\zeta)^2\right]. \tag{89}$$

Consequently,

$$R^{(1)} = -\frac{4}{a^2}\partial^2\zeta, \tag{90}$$

$$R^{(2)} = \frac{1}{a^2}\left[8\zeta\,\partial^2\zeta - 2(\partial\zeta)^2\right]. \tag{91}$$

13.2 Extrinsic curvature and the relational perturbation

Define

$$b \equiv \frac{\partial^2\beta}{a^2}. \tag{92}$$

Direct contraction of (86) gives the exact trace

$$K = \frac{1}{1+\alpha}\left[3(H+\dot{\zeta}) - \frac{e^{-2\zeta}}{a^2}\left(\partial^2\beta + \partial_i\zeta\,\partial_i\beta\right)\right]. \tag{93}$$

Writing

$$K = 3H + \delta K_1 + \delta K_2 + O(3), \tag{94}$$

one obtains

$$\delta K_1 = 3\dot{\zeta} - 3H\alpha - b, \tag{95}$$

$$\delta K_2 = 3H\alpha^2 - 3\alpha\dot{\zeta} + \alpha b - \frac{1}{a^2}\partial_i\zeta\,\partial_i\beta + \frac{2\zeta}{a^2}\partial^2\beta. \tag{96}$$

For a Fourier mode, let

$$u \equiv \dot{\zeta} - H\alpha, \qquad y \equiv \frac{k^2}{a^2}\beta. \tag{97}$$

Since $b = -y$, the linear relational perturbation becomes

$$\boxed{\delta K_1 = 3u + y}, \qquad \delta X_1 = \frac{3u+y}{3H_0}. \tag{98}$$

13.3 Trace–shear decomposition

Introduce the shear

$$\sigma_{ij} = K_{ij} - \frac{1}{3}K\gamma_{ij}. \tag{99}$$

The exact algebraic identity

$$K_{ij}K^{ij} - K^2 = \sigma_{ij}\sigma^{ij} - \frac{2}{3}K^2 \tag{100}$$

separates the trace response from the shift shear. Since the background shear vanishes, only its linear part is required at quadratic order:

$$\sigma_{ij}\sigma^{ij} = \frac{1}{a^4}\left[(\partial_i\partial_j\beta)^2 - \frac{1}{3}(\partial^2\beta)^2\right] + O(3). \tag{101}$$

For one Fourier mode this reduces to

$$\boxed{\sigma_{ij}\sigma^{ij} = \frac{2}{3}y^2}. \tag{102}$$

13.4 Equivalent scalar kernel of the RFG action

Define

$$F(X) \equiv -3X^2 Q(X) + V(X). \tag{103}$$

Using $K = 3H_0X$ and (100), the gravitational action is identically rewritten as

$$S_g = \frac{M_{\rm Pl}^2}{2} \int dt\, d^3x\, N\sqrt{\gamma} \left[Q(X)\, {}^{(3)}R + Q(X)\sigma_{ij}\sigma^{ij} + 2H_0^2 F(X) \right]. \tag{104}$$

Equation (104) is not an additional assumption; it is an algebraically equivalent form of the original RFG action.

Let

$$\bar{X} = \frac{H}{H_0}, \qquad \delta X_1 = \frac{\delta K_1}{3H_0}, \qquad \delta X_2 = \frac{\delta K_2}{3H_0}, \tag{105}$$

and evaluate Q, F and their derivatives at $\bar{X}$. Before use of the background equations or integrations by parts, the quadratic gravitational density is

$$\begin{aligned} \frac{\mathcal{L}_g^{(2)}}{(M_{\rm Pl}^2/2)a^3} &= QR^{(2)} + \left[Q(\alpha + 3\zeta) + Q_{,X}\delta X_1\right] R^{(1)} + Q\,\sigma_{ij}\sigma^{ij} \\ &\quad + 2H_0^2 \left[F\left(3\alpha\zeta + \frac{9}{2}\zeta^2\right) + F_{,X}\left(\delta X_2 + (\alpha + 3\zeta)\delta X_1\right) + \frac{1}{2}F_{,XX}\delta X_1^2 \right]. \end{aligned} \tag{106}$$

Proposition 13.3 (RFG Hessian identity)**.** *For*

$$Q(X) = 1 - \frac{\Omega_{R0}}{1+\theta} X^{-\theta}, \tag{107}$$

and the reconstructed potential of the homogeneous RFG branch,

$$F(X) = -3X^2 + \frac{3\Omega_{R0}}{1-\theta} X^{2-\theta}, \qquad (\theta \neq 1), \tag{108}$$

one has

$$F_{,XX} = -3D, \qquad D \equiv 2 - (2-\theta)\Omega_R, \qquad \Omega_R \equiv \Omega_{R0}X^{-\theta}. \tag{109}$$

Proof. Twice differentiating (108) gives

$$F_{,XX} = -6 + 3(2-\theta)\Omega_{R0}X^{-\theta} = -3\left[2 - (2-\theta)\Omega_R\right].$$

□

The trace-Hessian contribution to the Fourier-space quadratic kernel therefore takes the exact form

$$2H_0^2 \left(\frac{1}{2} F_{,XX} \delta X_1^2 \right) = -\frac{D}{3}(3u + y)^2, \tag{110}$$

whereas the shear contribution is

$$Q\,\sigma_{ij}\sigma^{ij} = \frac{2Q}{3} y^2. \tag{111}$$

Thus the lapse–shift kernel contains

$$\boxed{-\frac{D}{3}(3u+y)^2 + \frac{2Q}{3}y^2} \tag{112}$$

together with the curvature couplings displayed explicitly in (106).

13.5 Regular branch and the case $\theta = 1$

On the physical branch

$$0 \le \Omega_R < 1, \qquad \theta > 0, \tag{113}$$

the coefficients obey

$$D = 2(1 - \Omega_R) + \theta\Omega_R > 0, \qquad Q = 1 - \frac{\Omega_R}{1+\theta} > \frac{\theta}{1+\theta} > 0. \tag{114}$$

The trace and shear Hessians are therefore nondegenerate on this branch.

At $\theta = 1$, the regular logarithmic representative of the reconstructed potential yields, up to an irrelevant linear term,

$$F_{\theta=1}(X) = -3X^2 + 3\Omega_{R0} X \ln X. \tag{115}$$

Its second derivative is

$$F_{,XX} = -6 + \frac{3\Omega_{R0}}{X} = -3(2 - \Omega_R). \tag{116}$$

Hence (109) continues smoothly to

$$D = 2 - \Omega_R, \tag{117}$$

and $\theta = 1$ is not a degeneracy of the quadratic scalar kernel.

Remark 13.4 (Status of the scalar degree-of-freedom count)**.** *Equations (106)–(112) give the unreduced quadratic gravitational action. The pure-gravity FLRW lapse–shift reduction is now performed explicitly below. That calculation establishes that the pure-gravity scalar principal sector has no independent propagating $\dot\zeta^2$ term. It does not by itself determine the nonperturbative Dirac rank of the full metric–khronon theory, nor the matter-coupled scalar eigenvalues.*

14 Verified scalar constraint identities

The following identities have been verified directly from the quadratic expansion and are therefore recorded independently of the unresolved Hamiltonian reduction.

Lemma 14.1 (Second derivative identity)**.** *Defining*

$$D \equiv 2 - (2 - \theta)\Omega_R,$$

the Hessian of the effective potential satisfies

$$F_{XX} = -3D.$$

Proposition 14.2 (Quadratic trace–shear kernel)**.** *Introducing*

$$u = \dot\zeta - H\alpha, \qquad y = \frac{k^2}{a^2}\beta,$$

the quadratic kinetic kernel decomposes as

$$-\frac{D}{3}(3u + y)^2 + \frac{2}{3}Q\, y^2.$$

Hence the trace and shear sectors separate explicitly.

Lemma 14.3 (F_X block)**.** *Using the explicit second-order expansion of K and discarding only spatial boundary terms,*

$$\delta X_2 + (\alpha + 3\zeta)\delta X_1 \mathrel{\widehat{=}} \frac{3}{H_0}\zeta\, u.$$

Consequently the F_X contribution carries no independent dependence on the shift variable and therefore does not contribute to the momentum constraint after spatial integration by parts.

Remark 14.4. *The identities above are the checks used in the exact reduction below. They are not separate assumptions.*

15 Exact pure-gravity FLRW scalar reduction

This section incorporates the complete algebraic reduction of the pure-gravity scalar kernel. It is the calculation that decides whether the FLRW quadratic scalar sector contains an ordinary propagating gravitational scalar before matter is added.

Define

$$p \equiv \frac{k^2}{a^2}, \qquad N = 1+\alpha, \qquad N_i = \partial_i\beta, \qquad y \equiv p\beta, \qquad u \equiv \dot\zeta - H\alpha. \tag{118}$$

The scaled quadratic density is

$$L \equiv \frac{\mathcal{L}_g^{(2)}}{(M_{\rm Pl}^2/2)a^3}. \tag{119}$$

Keeping all terms contained in Eq. (106), the scalar density is

$$\begin{aligned} L = & -3DH^2\alpha^2 + 2DH\alpha y + 6DH\alpha\dot\zeta - \frac{D}{3}y^2 - 2Dy\dot\zeta - 3D\dot\zeta^2 \\ & + 6FH_0^2\alpha\zeta - 6F_X HH_0\alpha\zeta + 6F_X H_0\zeta\dot\zeta \\ & - \frac{4HQ_X}{H_0}\alpha p\zeta + 4Q\alpha p\zeta + \frac{2Q}{3}y^2 + \frac{4Q_X}{3H_0}py\zeta + \frac{4Q_X}{H_0}p\zeta\dot\zeta. \end{aligned} \tag{120}$$

Here D, Q, Q_X, F, F_X, H are functions of the homogeneous background and are held fixed during the algebraic lapse–shift elimination.

15.1 Lapse and shift constraints

Variation of Eq. (120) with respect to α gives

$$\boxed{0 = -3DH^2\alpha + DHy + 3DH\dot\zeta + 3FH_0^2\zeta - 3F_X HH_0\zeta - \frac{2HQ_X}{H_0}p\zeta + 2Qp\zeta.} \tag{121}$$

Variation with respect to y gives

$$\boxed{0 = -3DH\alpha + Dy + 3D\dot\zeta - 2Qy - \frac{2Q_X}{H_0}p\zeta.} \tag{122}$$

The Hessian in the nondynamical variables (α, y) is

$$\mathcal{H}_{\alpha y} = \begin{pmatrix} -6DH^2 & 2DH \\ 2DH & \frac{2}{3}(-D+2Q) \end{pmatrix}, \qquad \boxed{\det \mathcal{H}_{\alpha y} = -8DH^2Q.} \tag{123}$$

On the regular homogeneous branch,

$$D = 2(1-\Omega_R) + \theta\Omega_R > 0, \qquad Q = 1 - \frac{\Omega_R}{1+\theta} > \frac{\theta}{1+\theta} > 0,$$

so the algebraic system is invertible for $H \neq 0$.

15.2 Exact solution for the nondynamical variables

Solving Eqs. (121) and (122) gives

$$\boxed{y = \frac{\zeta\left(-3FH_0^2 + 3F_X HH_0 - 2Qp\right)}{2HQ}.} \tag{124}$$

The lapse perturbation is

$$\boxed{\begin{aligned}\alpha = \frac{\dot\zeta}{H} - \frac{p\zeta}{3H^2} - \frac{FH_0^2\zeta}{2H^2Q} + \frac{F_XH_0\zeta}{2HQ} + \frac{FH_0^2\zeta}{DH^2} - \frac{F_XH_0\zeta}{DH} \\ - \frac{2Q_Xp\zeta}{3DHH_0} + \frac{2Qp\zeta}{3DH^2}.\end{aligned}} \tag{125}$$

The leading cancellation is explicit: the principal part of the lapse is $\alpha_{\text{principal}} = \dot\zeta/H$, while the scalar shift has no principal time-derivative component.

15.3 Reduced density before integration by parts

Substitution of Eqs. (124) and (125) into Eq. (120) cancels every $\dot\zeta^2$ term exactly. The full reduced pure-gravity scalar density has the form

$$\boxed{L_{\text{red}} = A(t,p)\,\zeta\dot\zeta + B(t,p)\,\zeta^2,} \tag{126}$$

where

$$\boxed{A(t,p) = \frac{6FH_0^2 + 4Qp}{H}.} \tag{127}$$

The second coefficient is a polynomial in $p = k^2/a^2$:

$$\boxed{B(t,p) = B_0(t) + B_1(t)p + B_2(t)p^2,} \tag{128}$$

with

$$\boxed{B_0 = -\frac{3H_0^2(D-2Q)}{2DH^2Q}\,(FH_0 - F_XH)^2\,,} \tag{129}$$

$$\boxed{B_1 = -\frac{2\,(FH_0 - F_XH)\,(DH_0 + 2HQ_X - 2H_0Q)}{DH^2},} \tag{130}$$

and

$$\boxed{B_2 = -\frac{2}{3DH^2H_0^2}\left(DH_0^2Q - 2H^2Q_X^2 + 4HH_0QQ_X - 2H_0^2Q^2\right).} \tag{131}$$

Equations (126)–(131) are the exact missing symbolic reduction of the pure-gravity FLRW scalar kernel. They include the lower-derivative curvature and F_X contributions; they are not merely the principal limit.

15.4 Integration by parts and final scalar operator

The reduced action is

$$S^{(2)}_{\text{red}} = \frac{M_{\text{Pl}}^2}{2}\int \mathrm{d}t\,\mathrm{d}^3k\,a^3\left[A(t,p)\zeta_k\dot\zeta_{-k} + B(t,p)\zeta_k\zeta_{-k}\right]. \tag{132}$$

Because

$$a^3A\zeta\dot\zeta = \frac{1}{2}a^3A\frac{\mathrm{d}}{\mathrm{d}t}\zeta^2,$$

discarding the temporal boundary term gives

$$\boxed{S^{(2)}_{\text{red,IBP}} = \frac{M_{\text{Pl}}^2}{2}\int \mathrm{d}t\,\mathrm{d}^3k\,a^3M(t,p)\,\zeta_k\zeta_{-k},} \tag{133}$$

where

$$\boxed{M(t,p) = B(t,p) - \frac{1}{2}\left[\dot{A}(t,p) + 3HA(t,p)\right].} \tag{134}$$

Since $p = k^2/a^2$, one has $\dot{p} = -2Hp$, and therefore

$$\boxed{\dot{A} = 6H_0^2\left(\frac{\dot{F}}{H} - \frac{F\dot{H}}{H^2}\right) + 4p\left(\frac{\dot{Q}}{H} - \frac{Q\dot{H}}{H^2} - 2Q\right).} \tag{135}$$

Thus

$$M(t,p) = M_0(t) + M_1(t)p + B_2(t)p^2. \tag{136}$$

The $p^2\zeta^2$ term is fourth order in space, but it is not a time-propagating scalar kinetic term.

15.5 Principal-limit check

Keeping only the trace–shear principal kernel,

$$L_{\text{principal}} = -\frac{D}{3}(3u+y)^2 + \frac{2Q}{3}y^2, \qquad u = \dot{\zeta} - H\alpha, \tag{137}$$

the constraints reduce to

$$0 = 2DH(-3H\alpha + y + 3\dot{\zeta}), \tag{138}$$

$$0 = -\frac{2}{3}[-3DH\alpha + Dy + 3D\dot{\zeta} - 2Qy]. \tag{139}$$

For $DQH \neq 0$ the unique solution is

$$\boxed{\alpha = \frac{\dot{\zeta}}{H}, \qquad y = 0,} \tag{140}$$

and therefore

$$\boxed{L_{\text{principal,red}} = 0.} \tag{141}$$

This reproduces the exact reduction because the full reduced density (126) contains no $\dot{\zeta}^2$ term.

Theorem 15.1 (Absence of a pure-gravity FLRW scalar kinetic wave)**.** *On the regular homogeneous branch with $D > 0$, $Q > 0$ and $H \neq 0$, the pure-gravity scalar perturbation variables α and $y = k^2\beta/a^2$ are algebraically eliminable. The exact reduced FLRW scalar action contains no independent $\dot{\zeta}^2$ term. After integration by parts it is algebraic in time and takes the operator form Eq.* (133)*.*

Proof. The determinant $\det \mathcal{H}_{\alpha y} = -8DH^2Q$ is nonzero on the stated branch, so Eqs. (121) and (122) possess the unique algebraic solution (124)–(125). Direct substitution gives Eq. (126); explicit collection of powers of $\dot{\zeta}$ shows that the coefficient of $\dot{\zeta}^2$ vanishes identically. The remaining time-derivative term is $A\zeta\dot{\zeta}$, which is reduced to Eq. (134) by integration by parts with the measure a^3. □

Corollary 15.2 (What this result does and does not prove)**.** *The pure-gravity quadratic FLRW scalar sector does not contain a standard hyperbolic scalar wave before matter is added. If $M(t,p) \neq 0$, the linearized pure-gravity scalar equation is algebraic,*

$$M(t,p)\zeta_k = 0,$$

so generically $\zeta_k = 0$ in the pure-gravity scalar sector. If $M(t,p)$ vanishes on a special branch or scale, the correct interpretation is an additional degeneracy, gauge redundancy or strong-coupling gate, not an ordinary healthy propagating scalar. The result does not by itself determine the nonperturbative Dirac degree-of-freedom count and does not replace the matter-coupled scalar reduction.

16 Equation-by-equation derivation audit

The manuscript uses the following provenance rule: no displayed equation is allowed to stand without an origin. The origin classes are definition, axiom, variational equation, theorem, branch reconstruction, or open completion gate.

Table 4: Derivation map for the central equations. "Derived" means obtained algebraically or variationally from earlier displayed equations under the assumptions listed in the same row. "Gate" means that the equation is a precise target for a calculation not yet completed in this manuscript.

Equation or sector	Origin	Status
$\mathcal{C} > 0$, $\varphi = \ln(\mathcal{C}/\mathcal{C}_\star)$	Primitive ontology	Definition.
RCD action $S_{\rm RCD}$	Variational postulate	Fundamental scalar-sector definition.
RCD field equation	$\delta S_{\rm RCD}/\delta\varphi = 0$	Derived.
RCD Hamiltonian density	Legendre transform	Derived; bounded below for $\rho_\mathcal{C}, \kappa_\mathcal{C} > 0$ and $U \geq 0$.
RCD Noether currents	Time and space translations	Derived.
Finite-response equation	Constitutive relaxation law	Derived after specifying relaxation time and entropy functional; not microscopic.
RFG four-dimensional action	Metric–khronon foliation principle	Fundamental gravitational-sector definition.
$\Omega_R(X) = \Omega_{R0} X^{-\theta}$	Homogeneous branch law	Branch postulate connecting capacity scaling to foliation expansion.
$Q(X)$	Tensor normalization and paired response	Reconstructed.
$V(X)$	FLRW lapse variation and required branch equation	Reconstructed by theorem.
Background equation $E^2 = S + \Omega_{R0} E^{2-\theta}$	FLRW reduction of RFG action	Derived after branch law.
Unique expanding root	Global monotonicity theorem	Derived.
$w_R(a)$, $q(a)$, acceleration condition	Differentiation of the background equation	Derived.
Tensor action, Q_T, $c_T = 1$	Quadratic TT expansion	Derived in tensor sector.
Standard-siren ratio	WKB tensor amplitude transport	Derived under assumptions stated below Eq. (25).
Pure-gravity FLRW scalar reduction	Lapse–shift variation of Eq. (120)	Derived; $L_{\rm red} = A\zeta\dot{\zeta} + B\zeta^2$, no $\dot{\zeta}^2$, and $S^{(2)}_{\rm red,IBP}$ is governed by $M(t,p)$.
Matter-coupled scalar K, G, M matrices	Quadratic reduction after adding a matter scalar or fluid	Gate; the pure-gravity result does not determine the matter-coupled kinetic eigenvalues.
RFG-R one-field scalar K, G_1, G_2	Exact Schur complement of the RFG-R ADM action plus a canonical scalar	Derived for the stated comoving-gauge one-field system; the GR normalization and the regular reference branch are checked in Sec. 12.4. This does not establish a multi-fluid result.

Table 5: Derivation map for the central equations (continued).

Equation or sector	Origin	Status
RFG-R photon–baryon–CDM–neutrino core	Exact Sorkin–Schutz Schur reduction plus kinetic theory	Derived in Sec. 12.3.4; exact lapse–shift determinant, rational GR rank test, RFG-R inertia audit, photon polarization and neutrino hierarchy. No data likelihood is implied.
Nonperturbative Dirac rank	Canonical constraint algebra	Partially derived: the metric Legendre rank and the local Hamiltonian-constraint bracket are explicit. The global kernel, boundary-condition dependence, possible zero modes, and complete first-/second-class classification remain open.
PPN and screening	Static weak-field solution	Gate.

17 Absolute completion protocol

The present proof version is deliberately conservative. It separates a closed mathematical *definition* of an EFT from a completed empirical *test* of that EFT. RFG-R now contains its regular cosmological action, explicitly matched local-GR domain, exact finite physical multi-fluid constraint reduction, and untruncated kinetic hierarchy. It does not convert an unrun CMB likelihood, an unimplemented Einstein–Boltzmann solver, or the unresolved original-RFG Dirac problem into a result. A future version may be called a complete fundamental theory only if every remaining gate below is closed by equations and evaluated data likelihoods, not by wording.

17.1 Gate A: common microscopic action

The first missing derivation is the common origin of RCD and RFG. The required object is a single action

$$S_{\rm micro}[\Psi, g_{\mu\nu}, T]$$

whose coarse-grained macroscopic variable is $\mathcal{C}$ and whose homogeneous gravitational projection is $X = K/(3H_0)$. The required derivation is

$$S_{\rm micro} \longrightarrow \mathcal{C} \longrightarrow \Omega_R(X) = \Omega_{R0} X^{-\theta}.$$

Until this chain is supplied, the capacity-scaling law is a branch postulate and the theory is closed only conditionally on that postulate.

17.2 Gate B: pure-gravity FLRW scalar reduction

The pure-gravity FLRW scalar reduction is now completed in Sec. 15. The second variation of Eq. (15) around a scalar-perturbed FLRW metric and khronon,

$$N = 1 + \alpha_N, \qquad N_i = \partial_i \beta, \qquad \gamma_{ij} = a^2 e^{2\zeta} \delta_{ij}, \qquad T = t + \pi.$$

is reduced after varying with respect to the nondynamical fields. The completed pure-gravity result is

$$S^{(2)}_{\rm red,IBP} = \frac{M_{\rm Pl}^2}{2} \int {\rm d}t\, {\rm d}^3k\, a^3 M(t, k^2/a^2) \zeta_k \zeta_{-k},$$

with no independent $\dot{\zeta}^2$ term. Thus the pure-gravity scalar gate is not an ordinary Q_s, c_s^2 positivity problem; it is an algebraic constraint/degeneracy problem.

17.3 Gate B2: matter-coupled scalar reduction

For the original RFG action, the matter-coupled scalar calculation remains open. The pure-gravity degeneracy cannot be resolved by importing a result from a different EFT. For RFG-R, Sec. 12.3.4 now gives the exact physical photon–baryon–CDM–neutrino constraint reduction, while Sec. 12.4 supplies the one-canonical-scalar check. The general matter-coupled reduced action has the matrix form

$$S^{(2)}_{\mathrm{red,m}} = \frac{1}{2}\int \mathrm{d}t\,\mathrm{d}^3k\,a^3\left[\dot{\mathbf{q}}^T K\dot{\mathbf{q}} - \frac{k^2}{a^2}\mathbf{q}^T G\mathbf{q} - \mathbf{q}^T M\mathbf{q}\right],$$

followed by

$$\mathrm{eig}(K) > 0, \qquad \det\left(c_s^2 K - G\right) = 0, \qquad c_{s,i}^2 > 0.$$

This multi-component calculation is not implied by the pure-gravity result because matter sources the lapse and shift constraints. For RFG-R its finite constraint matrix and its exact kinetic hierarchy are now explicit; an actual spectrum calculation must still solve that differential–algebraic system together with recombination and test the resulting gradients. None of this settles the original RFG scalar degeneracy.

17.4 Gate C: Dirac constraint count

The Hamiltonian completion must distinguish the gauge-fixed preferred-foliation formulation from the fully covariant metric–khronon formulation. The local $\mathcal{C}$–$\mathcal{C}$ block is computed explicitly in Sec. 19; what remains is the completion of the primary, secondary and, if generated, tertiary constraints, together with the remaining Poisson-bracket blocks. In a fully covariant analysis this must be done before gauge fixing and must include the khronon canonical pair. The required output is the ranked Poisson-bracket matrix

$$\mathcal{C}_{AB}(\mathbf{x},\mathbf{y}) = \{\Phi_A(\mathbf{x}), \Phi_B(\mathbf{y})\},$$

the number of first- and second-class constraints, and the physical phase-space dimension

$$N_{\mathrm{dof}} = \frac{1}{2}\left(N_{\mathrm{phase}} - 2N_{\mathrm{first}} - N_{\mathrm{second}}\right).$$

Only after this count is complete can the scalar gravitational degree of freedom be certified as absent, healthy, or pathological.

17.5 Gate D: local gravity

For the original RFG action, the local limit cannot be inferred from the homogeneous background. It would require the static weak-field parametrization

$$\mathrm{d}s^2 = -(1+2\Phi)\mathrm{d}t^2 + (1-2\Psi)\delta_{ij}\mathrm{d}x^i\mathrm{d}x^j, \qquad T = t + \pi(r),$$

and direct solution of the field equations generated by Eq. (15). RFG-R instead gives a different, explicit low-energy model with the matching action in Eq. (37). In its local $s = 0$ domain the complete action is exactly GR, which is sufficient to derive the PPN prediction in Eq. (39); the packaged calculation and Cassini factor are reproducible from `PPN script`. This is a defined EFT matching rule, not a proof that the original singular RFG action has an independently derived screening solution.

The direct-original-RFG calculation would still return

$$\gamma_{\rm PPN} = \frac{\Psi}{\Phi}, \qquad \beta_{\rm PPN}, \qquad \alpha_1, \qquad \alpha_2,$$

and a derived screening or suppression mechanism. It is not used to justify the RFG-R PPN statement.

17.6 Gate E: Boltzmann and CMB completion

The cosmological background is not a CMB theory. RFG-R supplies the action derivatives, the exact extended-EFT map in Eq. (45), the sourced photon–baryon–CDM–neutrino reduction in Sec. 12.3.4, parameter set, rejection criteria, likelihood factors, and an exact input/output contract in `likelihood protocol`. The nonzero operator in Eq. (46) is absent from the public stock H-EFTCAMB interface, so that code is a validated GR reference only, not an exact RFG-R Boltzmann implementation. Moreover, the pure-gravity extended-EFT audit in Sec. 12.3.2 finds a degenerate $\dot{\zeta}^2$ coefficient and the original RFG-R DAE coefficient vanishes on a finite-wavenumber surface. RFG-RΞ in Sec. 12.3.6 is a different action that clears the present finite DAE root audit; it does not inherit a spectrum or likelihood from RFG-R. A complete empirical evaluation of that new action must derive and implement its own differential–algebraic hierarchy, retain all action-induced operators, and compute recombination, the drag epoch and sound horizon from the same action derivatives. The required extended Boltzmann implementation produces

$$C_\ell^{TT}, \quad C_\ell^{TE}, \quad C_\ell^{EE}, \quad C_\ell^{\phi\phi}, \quad P(k,z), \quad r_d,$$

and evaluates Eq. (50) with the stated public data. Until that compiled module and joint likelihood are actually run, RFG-RΞ is a derived candidate action, not an established replacement for the full standard cosmological pipeline.

18 Explicit open problems

The following sectors are not claimed to be closed by the present work:

Open problem 18.1 (Matter-coupled scalar perturbations of RFG)**.** *The pure-gravity FLRW scalar reduction is completed in Sec. 15. The remaining scalar perturbation problem is the matter-coupled reduction. Add a canonical scalar field, perfect fluid, or dust sector; vary the combined quadratic action with respect to α and β; solve the sourced constraints; and extract the reduced kinetic and gradient matrices K and G. The required viability conditions are*

$$\operatorname{eig}(K) > 0, \qquad \det\left(c_s^2 K - G\right) = 0, \qquad c_{s,i}^2 > 0.$$

Open problem 18.2 (Nonperturbative Dirac rank)**.** *The absence of a $\dot{\zeta}^2$ term in the pure-gravity FLRW scalar reduction does not by itself prove the full nonperturbative degree-of-freedom count. The metric Legendre rank and the local $\mathcal{C}$–$\mathcal{C}$ block are already explicit in Sec. 19. Complete the remaining canonical ADM/khronon analysis, state whether it is performed in unitary gauge or in the fully covariant formulation, construct the full Dirac matrix $\Delta_{AB} = \{\Phi_A, \Phi_B\}$, determine its global rank for specified boundary conditions, and only then compute $N_{\rm dof}$.*

Open problem 18.3 (RFG-R matter/CMB implementation and data inference)**.** *The exact RFG-R finite photon–baryon–CDM–neutrino constraint reduction is supplied in Sec. 12.3.4, alongside the exact massless and massive neutrino kinetic equations; the exact action-to-extended-EFT map is in Sec. 12.3.1. The pure-gravity audit in Sec. 12.3.2 still prevents a one-scalar backend. Implement the action-defined hierarchy with the nonzero coefficient $\bar{m}_5$ in Eq. (46);*

reproduce the GR limit; test the integrated gradient/stability conditions; then run the public likelihood in Eq. (50)*. The completion criterion is not a background fit: it is a released chain yielding CMB temperature, polarization and lensing spectra, matter power, growth, standard-siren predictions, stability flags, posterior samples, and a model-comparison statistic against* Λ*CDM and standard dark-energy competitors.*

Open problem 18.4 (Compatibility of matter coupling)**.** *Establish the compatibility between the assumed universal minimal metric coupling in RFG and the capacity-dependent clock and particle sectors of RCD. In particular, derive a unified particle action, identify the operational relation between ordering time and metric proper time, and test whether inertial and capacity responses remain universal.*

Open problem 18.5 (UV origin of the RFG-R matching rule)**.** *RFG-R is a low-energy multiscale EFT with an explicit local-GR matching rule. Derive that rule, or a more fundamental replacement for it, from a microscopic theory with a stated cutoff and matching conditions. The present local PPN prediction is operationally definite without that derivation, but its ultraviolet origin is not claimed.*

Open problem 18.6 (Quantum-relational completion)**.** *Formulate the microscopic relational microstates, spin, gauge structure and Born-rule sector.*

19 Canonical audit: exact results and unresolved closure conditions

This section supersedes any stronger claim of a completed Dirac algebra or of a numerically determined strong-coupling scale. The calculations below are exact for the metric Legendre map and for the local Hamiltonian-constraint bracket. The final degree-of-freedom count remains conditional on the global kernel of the resulting differential operator, the imposed boundary conditions, possible zero modes, and completion of the remaining Dirac algorithm.

19.1 Exact canonical momentum and kinetic Hessian

Define

$$\mathcal{U} \equiv {}^{(3)}R + K_{ij}K^{ij} - K^2, \qquad X = \frac{K}{3H_0}. \tag{142}$$

The momentum conjugate to γ_{ij} is

$$\boxed{\pi^{ij} = \frac{M_{\rm Pl}^2\sqrt{\gamma}}{2}\left[Q\left(K^{ij} - K\gamma^{ij}\right) + \frac{Q_{,X}}{6H_0}\mathcal{U}\,\gamma^{ij} + \frac{H_0}{3}V_{,X}\gamma^{ij}\right].} \tag{143}$$

Moreover $\pi_N \approx 0$ and $\pi_i \approx 0$. With $K_{ij} = \sigma_{ij} + K\gamma_{ij}/3$, the traceless block is

$$\pi_{\rm T}^{ij} = \frac{M_{\rm Pl}^2\sqrt{\gamma}}{2}Q\,\sigma^{ij}, \tag{144}$$

and the trace satisfies

$$\frac{2\pi}{M_{\rm Pl}^2\sqrt{\gamma}} = -2QK + \frac{Q_{,X}}{2H_0}\mathcal{U} + H_0V_{,X}. \tag{145}$$

At fixed $(\gamma_{ij}, \sigma_{ij})$,

$$\mathcal{W} = -2Q - \frac{4K}{3H_0}Q_{,X} + \frac{\mathcal{U}}{6H_0^2}Q_{,XX} + \frac{1}{3}V_{,XX}. \tag{146}$$

Hence the six-dimensional metric Legendre map is locally invertible iff

$$\boxed{Q \neq 0, \qquad \mathcal{W} \neq 0.} \tag{147}$$

On flat FLRW,

$$\mathcal{W}_{\rm FLRW} = -D, \qquad D = 2 - (2-\theta)\Omega_R, \tag{148}$$

so the physical branch $Q > 0, D > 0$ has no additional primary metric constraint. This is a completed result.

19.2 Explicit Hamiltonian-constraint bracket

The following calculation is performed in the preferred-foliation (unitary-gauge) ADM formulation, in which the khronon defines the time slicing. It is therefore a canonical analysis of the gauge-fixed spatially covariant theory, not yet the full covariant metric–khronon Dirac analysis with an independent canonical pair (T, p_T).

After the local inversion $K_{ij} = K_{ij}(\gamma, \pi)$, the canonical Hamiltonian and the total Hamiltonian are

$$H_C = \int d^3x \, [N\mathcal{C} + N^i\mathcal{H}_i], \tag{149}$$

$$H_T = \int d^3x \, \left[N\mathcal{C} + N^i\mathcal{H}_i + \lambda_N\pi_N + \lambda^i\pi_i\right], \qquad \mathcal{H}_i = -2\gamma_{ik}D_j\pi^{jk} + \mathcal{H}_{i,m}. \tag{150}$$

The Legendre transform gives the exact local identities

$$\frac{\delta\mathcal{C}[f]}{\delta\pi^{ij}} = 2fK_{ij}, \qquad \left.\frac{\partial\mathcal{C}}{\partial\,{}^{(3)}R}\right|_{\gamma,\pi} = -\frac{M_{\rm Pl}^2\sqrt{\gamma}}{2}\,Q. \tag{151}$$

The second identity follows from the envelope theorem: terms generated by the implicit variation of $K_{ij}(\gamma, \pi)$ cancel against the momentum term in the Legendre transform.

Let

$$P^{ij} \equiv K^{ij} - K\gamma^{ij}. \tag{152}$$

All ultralocal terms cancel under antisymmetrization of the two smearings. The only nontrivial contribution comes from the variation of the spatial Ricci scalar. Using

$$\delta\int d^3x\,\sqrt{\gamma}\,F\,{}^{(3)}R = \int d^3x\,\sqrt{\gamma}\left[-F\,{}^{(3)}G^{ij} + D^iD^jF - \gamma^{ij}D^2F\right]\delta\gamma_{ij}, \tag{153}$$

one obtains

$$\begin{aligned}
\{\mathcal{C}[f], \mathcal{C}[g]\} &= -M_{\rm Pl}^2\int d^3x\,\sqrt{\gamma}\left[gP^{ij}D_iD_j(fQ) - fP^{ij}D_iD_j(gQ)\right] \\
&= M_{\rm Pl}^2\int d^3x\,\sqrt{\gamma}\,(fD_ig - gD_if)\,\mathcal{V}^i,
\end{aligned} \tag{154}$$

where

$$\boxed{\mathcal{V}^i = P^{ij}D_jQ - QD_jP^{ij}.} \tag{155}$$

Equation (154) is the explicit local Poisson bracket, not merely its defining functional identity.

Preservation of $\mathcal{C}(x) \approx 0$ gives, modulo the momentum constraint, the first-order lapse equation

$$\boxed{2\mathcal{V}^iD_iN + ND_i\mathcal{V}^i \approx 0.} \tag{156}$$

Thus the rank question is equivalent to the kernel of the transport operator

$$\mathcal{L}_{\mathcal{V}}[N] \equiv 2\mathcal{V}^iD_iN + ND_i\mathcal{V}^i \tag{157}$$

with specified boundary conditions. Locally, wherever $\mathcal{V}^i \neq 0$, Eq. (156) constrains the lapse along the integral curves of $\mathcal{V}^i$, subject to compatibility, regularity and boundary data. Zeros of $\mathcal{V}^i$, closed integral curves, nontrivial topology and admissible zero modes can change the global kernel and therefore the rank.

On exact flat FLRW,

$$D_i Q = 0, \qquad D_j P^{ij} = 0, \qquad \Longrightarrow \qquad \mathcal{V}^i = 0, \tag{158}$$

and therefore

$$\{\mathcal{C}[f], \mathcal{C}[g]\}_{\mathrm{FLRW}} = 0. \tag{159}$$

The Hamiltonian-constraint operator loses rank identically on the FLRW background. This provides a nonperturbative explanation of why the FLRW scalar quadratic action cannot by itself determine the generic nonlinear number of degrees of freedom.

The exact conclusion is therefore:

$$\boxed{\begin{aligned} Q \neq 0,\ \mathcal{W} \neq 0 \ &\Longrightarrow \ \text{no primary metric degeneracy}, \\ \mathcal{V}^i_{\mathrm{FLRW}} = 0 \ &\Longrightarrow \ \text{Dirac-rank degeneracy on FLRW}. \end{aligned}} \tag{160}$$

A single background-independent integer N_{dof} additionally requires the global kernel of $\mathcal{L}_{\mathcal{V}}$, including boundary conditions and possible zero modes. It is not fixed by the homogeneous sector alone.

19.3 Cubic scalar sector and strong coupling

The exact quadratic FLRW reduction gives

$$S^{(2)}_{\mathrm{red}} = \int dt\, d^3k\, a^3 \mathcal{M}(t,k)\zeta^2, \qquad \frac{\partial^2 L^{(2)}_{\mathrm{red}}}{\partial \dot{\zeta}^2} = 0. \tag{161}$$

This proves loss of quadratic canonical normalization on exact FLRW. It does not by itself determine a strong-coupling scale. That scale requires the explicit reduced cubic action

$$S^{(3)}_{\mathrm{red}}[\zeta] = S^{(3)}[\alpha, \beta, \zeta]\big|_{\alpha=\alpha_{(1)}+\alpha_{(2)},\, \beta=\beta_{(1)}+\beta_{(2)}}, \tag{162}$$

followed by canonical normalization on a regulated background or in a specified higher-derivative completion. Therefore the rigorous conclusion is

$$\boxed{Q_s^{\mathrm{FLRW}} = 0, \qquad \Lambda_{\mathrm{sc}} \text{ is not determined by the quadratic action alone.}} \tag{163}$$

The stronger equation $\Lambda_{\mathrm{sc}} = 0$ is not retained as a derived result without the cubic coefficients and a precise limiting prescription.

19.4 Vacuum branch

For vanishing matter and radiation,

$$X^2 = \Omega_{R0} X^{2-\theta}, \tag{164}$$

so on the real positive branch

$$X_{\mathrm{vac}} = \Omega_{R0}^{1/\theta}, \qquad Q_{\mathrm{vac}} = \frac{\theta}{1+\theta}, \qquad D_{\mathrm{vac}} = \theta. \tag{165}$$

Thus the homogeneous de Sitter branch is regular with respect to the metric Legendre map for $\theta > 0$. Its full scalar stability, characteristic structure and nonlinear predictivity remain contingent on the unfinished Dirac and cubic analyses above.

19.5 Audited status

The calculations completed in this manuscript are:

$$\boxed{\begin{aligned} &\pi^{ij} \text{ is exact,} \\ &\operatorname{rank}\left(\frac{\partial \pi^{ij}}{\partial K_{kl}}\right) = 6 \quad \text{if} \quad Q \neq 0,\ \mathcal{W} \neq 0, \\ &\mathcal{W}_{\mathrm{FLRW}} = -D \quad \Longrightarrow \quad \operatorname{rank} = 6 \quad \text{on the FLRW branch } Q > 0,\ D > 0, \\ &\{\mathcal{C}[f], \mathcal{C}[g]\} \text{ is explicit,} \qquad Q_s^{\mathrm{FLRW}} = 0. \end{aligned}} \tag{166}$$

The following are not yet completed and must not be presented as proved:

$$\boxed{\begin{aligned} &\ker \mathcal{L}_{\mathcal{V}} \text{ for specified global boundary data,} \\ &N_{\mathrm{dof}} \text{ as a background-independent theorem,} \qquad S^{(3)}_{\mathrm{red}}, \qquad \Lambda_{\mathrm{sc}}. \end{aligned}} \tag{167}$$

20 Covariant KGB realization, executed likelihood, and production evidence

The preferred-foliation actions above remain distinct constructions with their own stated closure conditions. This section adds the covariant, luminal Kinetic Gravity Braiding (KGB) realization used for the native Einstein–Boltzmann calculation. It realizes the R-Universe background law with a standard scalar–tensor variational problem; it is not an algebraic replacement of the RFG-R or RFG-RΞ actions.

20.1 Action, field content, and global background definition

The action is the luminal KGB subclass of Horndeski theory [1, 2],

$$\begin{aligned} S_{\mathrm{RU}} &= \int \mathrm{d}^4x\sqrt{-g}\left[\frac{M_{\mathrm{Pl}}^2}{2}R + A(\phi)X + B(\phi)X^2 - V(\phi) - C(\phi)X\Box\phi\right] + S_m[g, \psi_m], \\ X &= -\frac{1}{2}\nabla_\mu\phi\nabla^\mu\phi. \end{aligned} \tag{168}$$

Thus $G_2 = AX + BX^2 - V$, $G_3 = -CX$, and $G_4 = M_{\mathrm{Pl}}^2/2$. All material fields are minimally coupled to the same Jordan metric. The constant Ricci coefficient gives, without a quasistatic or background approximation,

$$Q_T = M_{\mathrm{Pl}}^2, \qquad c_T^2 = 1, \qquad \frac{d_L^{\mathrm{GW}}}{d_L^{\mathrm{EM}}} = 1. \tag{169}$$

The action therefore has the two tensor polarizations and one scalar degree of freedom of this Horndeski subclass; it introduces neither an auxiliary khronon nor an ADM constraint substitution.

The functions A, B, C, V are analytic definitions through the scalar coordinate, not values interpolated from a finite background table. Set

$$\frac{\phi}{M_{\mathrm{Pl}}} = N = \ln a, \qquad a = \exp(\phi/M_{\mathrm{Pl}}), \tag{170}$$

and define $E = H/H_0$ as the unique positive root of

$$E^2 = \Omega_{m0}a^{-3} + \Omega_{r0}a^{-4} + r, \qquad r = \Omega_{R0}E^x, \qquad x = \alpha a_{\mathrm{eff}}(a). \tag{171}$$

At every finite a, this root is unique. Indeed, $0 \le x \le 2\alpha < 2$, and for $F(E) = E^2 - \Omega_{R0}E^x - \Omega_{m0}a^{-3} - \Omega_{r0}a^{-4}$,

$$F'(E) = E^{x-1}\left(2E^{2-x} - \Omega_{R0}x\right). \tag{172}$$

For $x > 0$, the parenthesis is strictly increasing from a negative value to infinity, so F first decreases and then increases; because $F(0) < 0$ and $F(E) \to \infty$, exactly one positive zero exists. For $x = 0$, F is strictly increasing for $E > 0$, with the same conclusion.

On the observed past light cone, $a \le 1$, take $a_{\text{eff}} = a$, so Eq. (171) is the fitted R-Universe R-alpha law. For a single-valued action at every real ϕ, let $a_{\text{eff}} = 2$ for $a \ge 2$, and on $1 < a < 2$ define the compact C^∞ window

$$s(t) = \frac{e^{-1/t}}{e^{-1/t} + e^{-1/(1-t)}}, \qquad t = a - 1, \qquad a_{\text{eff}} = a + (2-a)s(t). \tag{173}$$

All derivatives join smoothly at both endpoints. Since $2\alpha < 2$, the future limit is the finite de Sitter root

$$E_\infty = \Omega_{R0}^{1/(2-2\alpha)}. \tag{174}$$

The declared parameter point is

$$\Omega_{m0} = 0.3022734375, \quad \Omega_{r0} = 9.083909 \times 10^{-5}, \quad \Omega_{R0} = 0.69763572341, \quad \alpha = 0.497421875. \tag{175}$$

No additional cosmological parameter is fitted by the stability prescription below.

20.2 Exact reconstruction and scalar stability closure

Use a subscript N for $\mathrm{d}/\mathrm{d}N$. Differentiation of the root equation gives

$$E_N = \frac{r x_N \ln E - 3\Omega_{m0}a^{-3} - 4\Omega_{r0}a^{-4}}{2E - rx/E}, \qquad r_N = r\left(x_N \ln E + x\frac{E_N}{E}\right). \tag{176}$$

The denominator of Eq. (176) is directly tested over the declared global domain. In reduced $H_0 = M_{\text{Pl}} = 1$ reconstruction units, define

$$X = \frac{E^2}{2}, \qquad \rho_\phi = 3r, \qquad p_\phi = -3r - r_N, \qquad C = \frac{\alpha_B}{E^2}, \tag{177}$$

where the final equality fixes the braiding normalization in Eq. (168). Introduce

$$\begin{aligned} R_\rho &= \rho_\phi - 6E^2XC + 2C_\phi X^2, \\ R_p &= p_\phi + 2C_\phi X^2 + 2CXEE_N. \end{aligned} \tag{178}$$

The action functions are then fixed algebraically by

$$\begin{aligned} B &= \frac{\alpha_K E^2 - (R_\rho + R_p) + 8C_\phi X^2 - 12E^2XC}{8X^2}, \\ A &= \frac{R_\rho + R_p - 4BX^2}{2X}, \qquad V = \frac{R_\rho - R_p}{2} - BX^2. \end{aligned} \tag{179}$$

Here $C_\phi = C_N$ in the dimensionless scalar coordinate of Eq. (170). Direct substitution of Eq. (179) into the stress tensor of Eq. (168) returns ρ_ϕ and p_ϕ in Eq. (177); the homogeneous residual $\rho_{\phi,N} + 3(\rho_\phi + p_\phi)$ is therefore the scalar equation multiplied by the nonzero background $\dot\phi$.

Let $\Omega_i = \rho_i/(3M_{\rm Pl}^2 H^2)$. The braiding and scalar kinetic functions are fixed by

$$\begin{aligned}
\alpha_B &= \frac{\Omega_R}{3}, \qquad \alpha_{B,N} = \frac{\Omega_{R,N}}{3}, \\
\mathcal{H}_e &= 3\Omega_m + 4\Omega_r, \qquad -\frac{E_N}{E} = \frac{\mathcal{H}_e - \Omega_R x_N \ln E}{2 - \Omega_R x}, \\
\mathcal{N}_s^{(0)} &= \frac{\Omega_R \left(\mathcal{H}_e x - 2x_N \ln E\right)}{2 - \Omega_R x}, \\
\mathcal{N}_s &= \mathcal{N}_s^{(0)} + \alpha_B \left(1 + \frac{E_N}{E}\right) + \alpha_{B,N} - \frac{\alpha_B^2}{2}, \\
D &= \frac{\mathcal{N}_s}{c_\star^2}, \qquad \alpha_K = D - \frac{3}{2}\alpha_B^2, \qquad c_\star^2 = 1, \\
\frac{Q_s}{M_{\rm Pl}^2} &= \frac{2D}{(2-\alpha_B)^2}, \qquad c_s^2 = \frac{\mathcal{N}_s}{D} = 1.
\end{aligned} \tag{180}$$

This closure contains no fitted cosmological parameter. In radiation domination, $\Omega_{R,N} \to 4\Omega_R$ and $-E_N/E \to 2$, giving $D/\Omega_R \to 1$. Thus the coefficient $1/3$ is fixed by the requirement that the scalar kinetic sector decouple with the R-sector energy fraction, rather than leaving a finite early-time EFT coupling. The displayed form of $\mathcal{N}_s$ is algebraically equivalent to the Bellini–Sawicki expression but avoids cancellation of order-unity radiation terms. The nontrivial scalar stability test is $\mathcal{N}_s > 0$, which was scanned directly rather than inferred from the background equation.

Table 6: Executed action-level validation of the covariant KGB R-Universe realization on $10^{-8} \le a \le 10^3$.

Test	Result
Maximum relative background residual	6.179×10^{-16}
Maximum relative scalar-conservation residual	1.592×10^{-16}
Maximum relative reconstructed density residual	4.219×10^{-16}
Maximum relative reconstructed pressure residual	3.642×10^{-16}
Independent high-precision action residual	1.748×10^{-81}
$\min(\mathcal{N}_s, D, Q_s/M_{\rm Pl}^2)$	$(7.680, 7.680, 3.840) \times 10^{-29}$
$\max\lvert c_s^2 - 1\rvert$	0
Present $(w_R, \alpha_B, \alpha_K, Q_s/M_{\rm Pl}^2)$	$(-0.90900, 0.23255, 0.37898, 0.29457)$

The complete symbolic derivation and executable implementation are part of the paper's reproducibility record: `derivation`, `action constructor`, `precision validator`, `machine validation record`, and `global stability scan`. The independent scan over $\alpha \in [0.02, 0.95]$ and $\Omega_{m0} \in [0.24, 0.38]$ has positive global minima $\min(\mathcal{N}_s, D, Q_s/M_{\rm Pl}^2) = (6.824, 6.824, 3.412) \times 10^{-29}$.

20.3 Matter, pre-recombination, and local-gravity gates

The declared action also gives directly evaluated physical gates. Its quasistatic calculation gives

$$1 \le \mu(a) \le 1.05876714, \qquad 0.99939305 \le \frac{D_+(a)}{D_{+,\Lambda\rm CDM}(a)} \le 1.01046983 \tag{181}$$

on the stated background interval. The pre-recombination integration gives

$$\frac{r_s^{\rm KGB}}{r_s^{\Lambda\rm CDM}} - 1 = -4.898 \times 10^{-13}, \tag{182}$$

because the reconstructed response is negligible in that era. The local screening calculation gives a Vainshtein radius of 97.81 pc for the declared Solar source and a screened PPN envelope

$$|\gamma_{\rm PPN} - 1| \le 8.325 \times 10^{-16} < 2.3 \times 10^{-5} [22]. \tag{183}$$

The calculations and their input/output contract are public in `the matter calculation`, `the pre-recombination calculation`, `the screening calculation`, and `the Boltzmann contract`.

20.4 Native Einstein–Boltzmann and Planck 2018 calculation

The action in Eq. (168) was passed directly to the native reparametrized-Horndeski (RPH) interface of H–EFTCAMB[15]. The source generator `generates solver input` from the exact functions in `the action constructor`: $w_R(a)$, $\alpha_K(a)$, and $\alpha_B(a)$, with $\alpha_M = \alpha_T = 0$ identically. H–EFTCAMB uses the convention $\alpha_B^{\rm RPH} = -\alpha_B^{\rm BS}/2$; this conversion is explicit in the generator and does not alter the action convention above.

The native run evolves photons, baryons, CDM, three massless neutrino species ($N_{\rm eff} = 3.046$), metric constraints, and the KGB scalar, with recombination and lensing enabled. The fixed input point has $\omega_b = 0.02237$, $A_s = 2.1 \times 10^{-9}$, $n_s = 0.9649$, $\tau = 0.0544$, and $A_{\rm Planck} = 1$. The action's radiation fraction sets $H_0 = 67.86230797\,{\rm km\,s^{-1}\,Mpc^{-1}}$ and, together with the declared Ω_{m0}, fixes $\omega_c = 0.116836$. These are fixed inputs for this calculation, not inferred parameter values.

The outputs $C_\ell^{TT}, C_\ell^{TE}, C_\ell^{EE}, C_\ell^{\phi\phi}$ are evaluated by `the Planck wrapper` against the official Planck 2018 Plik-lite TTTEEE, Commander low-ℓ TT, SimAll low-ℓ EE, and CMB-dependent lensing likelihood objects [16]. CAMB output D_ℓ values are converted to raw C_ℓ before the official likelihood call; the lensing-potential column is separately converted with $D_L^{\phi\phi} = [L(L+1)]^2 C_L^{\phi\phi}/(2\pi)$. The likelihood library executes its own reference-vector self-test before each evaluation.

Table 7: Executed official Planck 2018 likelihood values at the declared fixed KGB R-Universe point. The absolute sum includes likelihood normalizations and is not a goodness-of-fit χ^2.

Component	$-2\ln\mathcal{L}$
Plik-lite TTTEEE, $2 \le \ell \le 2508$	747.6507901
Commander TT, $2 \le \ell \le 29$	23.0986580
SimAll EE, $2 \le \ell \le 29$	395.9799428
CMB-dependent lensing	43.4305271
Total	1210.1599180

The 601-node RPH spline is the reference numerical realization. Direct 201-versus-601-node recalculation changes the total by only 8.84×10^{-6} in $-2\ln\mathcal{L}$. Moving the numerical perturbation turn-on from $a = 10^{-4}$ to 10^{-3} changes it by -2.94×10^{-5}. The solver and likelihood audit records are `solver convergence`, `Planck summary`, and `the implementation contract`.

This closes a genuine CMB-plus-lensing likelihood evaluation for one specified KGB point. The conditional local profile in Sec. 20.4.1 adds a restricted diagnostic, but does not construct a posterior or optimize the full cosmological and nuisance-parameter space. The exact posterior in Sec. 20.5 is a separate four-chain inference for the declared Planck–Pantheon+–DESI DR2 BAO–chronometer probe set, while the production evidence comparison is reported in Sec. 20.7. The conditional calculation includes exact Pantheon+, DESI DR2 BAO, and chronometer likelihood ordinates; the native RSD calculation remains an audit rather than a survey likelihood. The

RCD-to-KGB identification in Eq. (170) remains the macroscopic branch chosen in this work; a common microscopic action deriving that identification is not represented as a completed result.

20.4.1 Conditional KGB–ΛCDM profile

The executed fixed-$A_{\rm Planck} = 1$ conditional surface contains 154 KGB points. Every point combines the official Planck 2018 ordinate with an exact-background Pantheon+ full-covariance relative-distance term, DESI DR2 BAO full-covariance term, and cosmic-chronometer term; the drag scale is read from the H–EFTCAMB log of that same point. The lowest sampled surface record is 2493.222580514448 at $\alpha = 0.100$, $\Omega_{m0} = 0.30750$. It is a finite conditional grid, not a continuous fit: all primordial, recombination, calibration, and high-ℓ nuisance inputs remain fixed.

A separate local refinement selects $\alpha = 0.0975$, $\Omega_{m0} = 0.3075$, and reconstructs and validates the complete KGB action at that point. The exact late-time integration uses the 1580-supernova Pantheon+ sample with its supplied full covariance and analytically marginalized intercept [17], 13 DESI DR2 BAO measurements with their full covariance [18], and the 31-point cosmic-chronometer compilation documented in Ref. [19]. The chronometer file supplies diagonal uncertainties only, which is the stated likelihood treatment. Its solver-derived value is $r_{\rm drag} = 147.32\,{\rm Mpc}$. The separately executed fixed-spectrum calibration grids let each matched model take its own $A_{\rm Planck}$ minimum. The Planck terms include their likelihood normalizations and are not goodness-of-fit χ^2 values.

Table 8: Executed matched fixed-input Planck plus late-time conditional record at $\alpha = 0.0975$, $\Omega_{m0} = 0.3075$. Each model uses its own one-dimensional fixed-spectrum $A_{\rm Planck}$ minimum; all other cosmological, primordial, recombination, and high-ℓ nuisance inputs are matched and fixed. This table is neither a posterior nor a model-selection statistic.

Calculation	α	Ω_{m0}	$A_{\rm Planck}$	Planck $-2\ln\mathcal{L}$	$\chi^2_{\rm late}$
KGB	0.0975	0.30750	1.00275	1063.6712515	1416.5609849
Matched standard-ΛCDM	0	0.30750	1.00325	1060.2194089	1422.5765928

The KGB sum is 2480.2322364, the matched standard-ΛCDM sum is 2482.7960016, and their recorded difference is

$$\left[-2\ln\mathcal{L}_{\rm Planck} + \chi^2_{\rm late}\right]_{\rm KGB} - \left[-2\ln\mathcal{L}_{\rm Planck} + \chi^2_{\rm late}\right]_{\Lambda{\rm CDM}} = -2.5637653. \tag{184}$$

This is an executed conditional result, not a common-nuisance fit, posterior, or Bayesian evidence. The action validation at this point has a maximum 80-digit reconstruction residual 2.076×10^{-81}; the separately evaluated cubic-KGB local screening gate gives $r_V = 102.56\,{\rm pc}$ and $|\gamma_{\rm PPN} - 1| \leq 1.0463 \times 10^{-15}$.

The same selected KGB point was also evolved by H–EFTCAMB at 58 transfer redshifts. Its linear total-matter $P(k, z)$ yields $\sigma_8(z)$ by direct top-hat integration and $f\sigma_8 = d\sigma_8/d\ln a$ by a symmetric finite difference. It covers all 23 entries of the local RSD compilation, composed of the internally vetted 22-point set and its documented added BOSS DR12 CMASS datum [20, 21]. The largest difference from H–EFTCAMB's independent velocity–density estimator is 9.385×10^{-5}, and halving the redshift difference shifts the largest prediction by only 5.009×10^{-5}. The compilation lacks survey covariance, window/AP mapping, and nonlinear nuisance modeling for its correlated groups, so it is an RSD residual audit and is intentionally not added to Eq. (184).

The complete action generator, Planck profiler, exact late-time evaluator, native RSD evaluator, selected-point PPN gate, and machine-readable ledger are public as `the profile generator`,

Table 9: Marginal KGB posterior summaries from the converged exact Planck–Pantheon+–DESI DR2 BAO–chronometer chain set. The interval is the equal-tail 68% credible interval after the declared burn-in removal.

Parameter	Median	68% interval	$\hat{R}$	bulk ESS	tail ESS
α	0.420905	$[0.276975, 0.579224]$	1.0023	823.9	1420.8
Ω_{m0}	0.306403	$[0.300921, 0.312209]$	1.0035	959.9	1577.5
Ω_{r0}	9.21285×10^{-5}	$[9.05209 \times 10^{-5}, 9.38543 \times 10^{-5}]$	1.0030	926.9	1489.3
ω_b	0.022495	$[0.022365, 0.022627]$	1.0056	690.2	1330.0
$\ln(10^{10} A_s)$	3.00121	$[2.98266, 3.01686]$	1.0035	944.2	1429.1
n_s	0.97077	$[0.967194, 0.974032]$	1.0029	779.4	1214.6
τ	0.0399769	$[0.0300871, 0.0475536]$	1.0034	938.2	1341.3
A_{Planck}	0.998996	$[0.996612, 1.0016]$	1.0066	965.0	1541.9

`the late-time evaluator`, `the RSD evaluator`, `the conditional ledger`, and `the scope statement`. The role of ΛCDM is an observational benchmark, not a proposed fundamental theory to retain alongside R-Universe. The specified KGB action is a physical replacement candidate. The exact posterior in Sec. 20.5 covers its stated Planck–Pantheon+–DESI DR2 BAO–chronometer probe set, and the production evidence in Sec. 20.7 completes the technical marginal-likelihood comparison for the same declared probe family. An official RSD likelihood and a broader matched model comparison remain required before empirical replacement can be asserted.

20.5 Exact joint posterior and convergence diagnostic

The covariant KGB realization was sampled with 4 independent exact chains. Here "exact" denotes direct evaluation of the declared numerical action–solver–likelihood pipeline rather than an emulator, response approximation, or surrogate; the documented numerical integration and solver tolerances still apply. Every likelihood evaluation regenerated the R-Universe KGB response, ran H–EFTCAMB, and evaluated Planck 2018 CMB plus lensing (including the Gaussian absolute-calibration factor for A_{Planck}), Pantheon+, DESI DR2 BAO, and cosmic-chronometer data. The RSD compilation is not included: the available rows do not supply the survey covariance, window, AP mapping, and nonlinear nuisance likelihood required for a valid RSD inference. A per-chain 25% burn-in is removed before forming the reported marginal intervals.

All rank-normalized, folded split-$\hat{R}$ diagnostics are below 1.01; the largest is $\hat{R} = 1.0066$. The rank-normalized bulk and 5%/95% tail ESS values are each at least 400 and 400, respectively; their minima are 690.2 and 1214.6. The retained sample set contains 29256 weighted MCMC states. The smallest retained value is $-2 \ln \mathcal{L} = 2451.92$; it is a sampled likelihood ordinate, not a Bayesian evidence or a model-selection statistic.

The calculation is reproducible from the saved chain files and the public chain runner, posterior analyzer, numerical summary, and hash-bound production audit named below. It establishes a converged posterior only for the explicitly stated KGB action and probe set. The separate Bayesian-evidence comparison with the local ΛCDM reference is reported in Sec. 20.7; neither calculation is an inference for the distinct RFG-R and RFG-RΞ actions.

20.6 ΛCDM control posterior and data-interface support

As an independent control on the same local inference environment, a ΛCDM reference posterior was executed with the same Planck 2018 high-ℓ Plik-lite TTTEEE, low-ℓ Commander and

Table 10: ΛCDM control posterior from four local 12,000-sample chains. The entries use a 30% per-chain burn-in and weighted retained samples. The control run is an empirical calibration of the data interface, not a constraint that the R-Universe KGB posterior must reproduce the same parameter values.

Parameter	Mean	68% interval	95% interval
Ω_{m0}	0.29707	$[0.29357, 0.30048]$	$[0.29034, 0.30399]$
Ω_{r0}	8.8645×10^{-5}	$[8.7914, 8.9330] \times 10^{-5}$	$[8.7241, 9.0060] \times 10^{-5}$
ω_b	0.022404	$[0.022288, 0.022521]$	$[0.022171, 0.022645]$
$\ln(10^{10}A_s)$	3.00201	$[2.98720, 3.01843]$	$[2.96532, 3.03390]$
n_s	0.96823	$[0.96497, 0.97177]$	$[0.96181, 0.97456]$
τ	0.03863	$[0.02957, 0.04690]$	$[0.01940, 0.05440]$
A_{Planck}	0.99925	$[0.99683, 1.00204]$	$[0.99430, 1.00413]$
χ^2	2461.50	$[2457.71, 2464.90]$	$[2456.22, 2470.33]$

SimAll, lensing, Pantheon+, DESI DR2 BAO, and cosmic-chronometer probe family used to define the R-Universe empirical boundary above. Four independent local chains, initialized with independent seeds, were continued to 12,000 accepted samples each. The run was not used to force the R-Universe KGB posterior onto the ΛCDM parameter surface. Its purpose is a stricter control: it verifies that the same data plumbing, likelihood normalization, nuisance treatment, and late-time distance pipeline recover a standard ΛCDM benchmark region when the model is reduced to the phenomenological reference case.

After discarding the first 30% of each local chain, the combined retained sample gives the marginal control summary in Table 10. The best sampled point has

$$\chi^2_{\min} = 2454.7832, \qquad -\ln \mathcal{P}_{\min} = 1203.1628,$$

at

$$\Omega_{m0} = 0.296278, \quad \Omega_{r0} = 8.8531 \times 10^{-5}, \quad \omega_b = 0.022420,$$
$$\ln\left(10^{10}A_s\right) = 3.00870, \quad n_s = 0.968625, \quad \tau = 0.042710,$$
$$A_{\text{Planck}} = 0.998985.$$

The split-chain convergence of this fast local control is adequate for a calibration statement but deliberately not presented as the same production standard as the released KGB posterior: the simple split-$\hat{R}$ values over the retained local chains lie approximately in the range 1.03–1.10. This is nevertheless a direct positive data check. The ΛCDM control lands in the expected CMB-plus-late-time region with a stable calibration amplitude $A_{\text{Planck}} \simeq 1$, while the R-Universe KGB posterior in Table 9 remains a separate action-defined inference. Agreement of the data interface with the standard benchmark therefore supports the empirical credibility of the R-Universe execution; lack of one-to-one equality in the fitted parameters is not a failure mode, because the R-Universe action is not a reparametrized copy of ΛCDM. This is evidence for the data-facing implementation in the ordinary empirical sense. It should not be relabelled as Bayesian evidence $Z = \int \mathcal{L}\pi \, d\theta$ without a dedicated marginal-likelihood calculation for the models being compared. That dedicated calculation has now been performed and is reported next.

20.7 Production Bayesian evidence

The final model-comparison calculation uses dynamic nested sampling over the explicit Cobaya priors stated for each model, with the same Planck 2018 CMB-plus-lensing, Pantheon+, DESI DR2 BAO, and cosmic-chronometer probe family used in the posterior analyses above. Both production runs use `nlive` = 250, `maxcall` = 20000, $\Delta \log Z_{\text{stop}} = 0.3$, the `multi` bound, and `rwalk` sampling. The resulting marginal likelihoods are

Table 11: Production dynamic nested-sampling evidence for the covariant KGB R-Universe realization and the local ΛCDM reference under their declared priors and the stated CMB-plus-late-time probe family. Larger $\log Z$ is favored.

Model	$\log Z$	$\sigma_{\log Z}$	$-2\log Z$	Calls	Status
Covariant KGB R-Universe	-1335.5996905	2.8042019	2671.1993809	20268	completed
Local ΛCDM reference	-1291.3199169	2.8135177	2582.6398339	20264	completed

Thus

$$\Delta \log Z \equiv \log Z_{\mathrm{KGB}} - \log Z_{\Lambda\mathrm{CDM}} = -44.2797735, \qquad -2\Delta \log Z = 88.5595471. \tag{185}$$

With the conventional evidence orientation, this production calculation favors the local ΛCDM reference over the present covariant KGB realization for the declared prior volumes and probe set. This does not invalidate the action construction, stability tests, fixed-point likelihood execution, conditional-profile improvement in Eq. (184), or converged KGB posterior. It does, however, close the previous evidence gate: the present KGB realization is not favored by the technical Bayesian marginal-likelihood comparison against the stated local ΛCDM control. The native RSD calculation remains outside this evidence because the available compilation is not a survey-complete likelihood.

Archive and production verification. The released posterior was independently audited over all 46206 cached exact likelihood points. The deep cache audit passed after verification of the archive SHA-256 values and independent member-by-member validation; the production-bundle audit also passed and is hash-bound to the numerical summary. These checks establish the integrity and reproducibility of the stated KGB posterior, not a model comparison or an RSD likelihood.

Versioned posterior archive. The immutable public record is release `v1.6.1`. It contains the `numerical summary`, the `production-bundle audit`, the `deep point-cache audit`, and the provenance-marked `TeX record`. The released chain files are `05`, `06`, `07`, and `08`. The archive also publishes the `chain runner`, `posterior analyzer`, and `read-only release verifier`.

21 What has been established

1. The conservative scalar sector of RCD is mathematically closed at the classical local-field level considered here: its action, variational field equation, Hamiltonian, Noether currents, linear stability and hyperbolicity are derived. The finite-response transport extension is closed only after specification of its free-energy functional and constitutive coefficients.

2. The homogeneous cosmology of RFG is closed within the reconstructed action and the stated matter assumptions. Existence and uniqueness of the expanding branch, the exact logarithmic Hubble derivative, deceleration, effective equation of state, acceleration–capacity identity and the de Sitter attractor are proved.

3. The tensor sector of RFG is closed, ghost-free on the homogeneous branch, luminal, and predicts a definite standard-siren distance ratio.

4. The pure-gravity FLRW scalar reduction is explicitly computed: lapse and scalar shift constraints are solved, the exact reduced density is $L_{\mathrm{red}} = A\zeta\dot{\zeta} + B\zeta^2$, the coefficient of $\dot{\zeta}^2$ vanishes identically, and the integration-by-parts form is governed by $M(t, k^2/a^2)$.

5. The nonperturbative metric kinetic rank and the local Hamiltonian-constraint bracket are explicit, but the full global Dirac rank is not. Since the scalar quadratic kinetic term vanishes on exact FLRW, the scalar sector exhibits a linearization degeneracy and lacks a standard quadratic canonical normalization. A strong-coupling scale cannot be assigned without the explicit reduced cubic action together with either a regulated background or a specified higher-derivative completion.

6. RFG-R is an explicit regular multiscale EFT. Its potential is reconstructed from its target background equation, its recovery of the original cosmological branch and regular $X \to 0$ limit are numerically tested, and its tensor normalization remains positive on the published grid. The versioned public release contains the source, released inputs, output tables, and validation records linked in Sec. 12; externally licensed likelihood products remain subject to their own distribution terms.

7. RFG-R has a defined local-GR domain. In that domain its PPN prediction is exactly the GR value and the included Cassini likelihood factor is reproducible. This is a property of the stated matching action, not an observational model-selection result.

8. For RFG-R plus one minimally coupled canonical scalar, the sourced lapse–shift system and exact reduced quadratic action are derived. The GR stiff-field normalization $K = 6$, $c_s^2 = 1$ is recovered, and the published reference branch has positive Δ, K, and G_1, with G_2 non-negative to numerical precision. This is a one-field internal check, not a physical multi-fluid cosmology.

9. The RFG-R action is exactly mapped into the extended EFT basis. The required $\bar{m}_5 \delta^{(3)} R \delta K$ coefficient is nonzero on the reference branch and has an independent numerical validation. The complete pure-gravity extended-EFT scalar audit retains that operator and finds a degenerate $\dot{\zeta}^2$ coefficient on all 2,401 reference (a, k) points. The public stock Boltzmann backend also lacks $\bar{m}_5$, so it supplies a GR reference calculation rather than an RFG-R CMB prediction.

10. The RFG-R photon–baryon–CDM–neutrino finite constraint action is explicitly reduced, with a rational-arithmetic GR rank check and a data-documented RFG-R reference inertia audit. Photon polarization and neutrino anisotropic stress remain in their exact kinetic hierarchies, including the unprojected massive-neutrino equation. The full matter/CMB likelihood has fixed data, spectra and rejection rules, but has not yet been evaluated; an action-faithful Einstein–Boltzmann solver and a data run are still required.

11. RFG-RΞ is a separately defined R-Universe action, obtained by direct quadratic variation of the background-null operator $\Xi({}^{(3)}R + \sigma_{ij}\sigma^{ij})$. It preserves the R-Universe homogeneous branch and $c_T = 1$. At the unfitted normalization benchmark $\Xi = 1$, the exact 49-by-81 multi-fluid DAE audit finds no curvature-constraint zero, with positive tensor and material kinetic diagnostics; the independent $\Xi = 2$ audit has the same no-root result. The locally matched Cassini factor is directly evaluated as $-2 \ln \mathcal{L}_{\mathrm{Cassini}} = 0.8336483932$ for both benchmarks, while a full CMB/matter likelihood remains uncomputed. This is finite-grid internal viability evidence, not a global stability proof or an observational fit.

12. The covariant KGB realization in Sec. 20 has a globally defined action, algebraically reconstructed functions, positive stated scalar-stability coefficients, and luminal tensor propagation. Its native H–EFTCAMB photon–baryon–CDM–neutrino evolution has an executed official Planck 2018 CMB-plus-lensing likelihood, an exact-background Pantheon+–DESI DR2 BAO–chronometer conditional record, and a selected-point local screening gate. At the stated matched fixed-input point, the KGB-minus-ΛCDM conditional sum is -2.5637653; the native 23-point RSD calculation is a converged residual audit, not a

likelihood. The exact posterior in Sec. 20.5 applies only to this KGB action and its declared probe set.

13. The local ΛCDM control posterior in Sec. 20.6 is a direct data-support check for the empirical pipeline. With the same CMB and late-time probe family, four independent 12,000-sample chains recover the standard benchmark region, $A_{\text{Planck}} \simeq 1$, and a best sampled $\chi^2 = 2454.7832$. This supports the R-Universe data execution by showing that the interface behaves correctly on the phenomenological reference model. It does not require the R-Universe KGB posterior to copy the same parameter values, because the KGB action is a distinct physical realization rather than a reparametrized ΛCDM surface.

14. The production Bayesian evidence calculation in Sec. 20.7 is complete for the stated KGB and local ΛCDM models under their declared priors and the same CMB-plus-late-time probe family. It gives $\log Z_{\text{KGB}} = -1335.5996905 \pm 2.8042019$ and $\log Z_{\Lambda\text{CDM}} = -1291.3199169 \pm 2.8135177$, or $\Delta \log Z = -44.2797735$. Therefore this technical marginal-likelihood comparison favors the local ΛCDM reference, even though the restricted fixed-input conditional profile slightly favors the selected KGB point. This evidence comparison is not an RFG-R or RFG-RΞ likelihood evaluation and still excludes RSD as a survey likelihood.

22 Conclusion

We have assembled a single, transparent theoretical document that contains the mathematically derived layers of Relational Capacity Dynamics and its preferred-foliation construction, plus a separate, explicit RFG-R effective completion. The scalar macroscopic theory is ghost-free, gradient-stable and hyperbolic under the stated conditions. The covariant branch produces a predictive homogeneous cosmology, a clean tensor sector with $c_T = 1$, and an exactly reduced pure-gravity FLRW scalar sector with no independent $\dot{\zeta}^2$ kinetic wave. RFG-R regularizes the low-X action, gives an unambiguous local-GR/PPN domain, and makes every stated numerical validation reproducible from the public source. Its exact one-canonical-scalar reduction provides a nontrivial sourced-constraint and GR-limit check on the regular reference branch. The action is also mapped directly into the extended EFT perturbation basis; that derivation fixes the nonzero $\bar{m}_5 \delta^{(3)} R \delta K$ term rather than treating it as a tunable omission.

The metric Legendre map is regular on the physical branch, while the full Dirac rank and the cubic scalar cutoff remain open. Exact FLRW has vanishing quadratic scalar kinetic normalization. For the regular RFG-R action, the complete pure-gravity extended-EFT audit likewise yields a degenerate $\dot{\zeta}^2$ coefficient despite a nonsingular lapse–shift determinant. The sourced photon–baryon–CDM–neutrino constraint reduction is now fully varied, retaining $\bar{m}_5$, the spatial trace and traceless equations, and the kinetic hierarchies. Its exact DAE closure rejects the original RFG-R reference branch because μ_ζ vanishes at finite wavenumber. The distinct background-preserving RFG-RΞ action in Sec. 12.3.6 removes that failure on its stated finite grid without changing the R-Universe Friedmann branch or $c_T = 1$. It is nevertheless a new theory, not an empirical comparison. An action-faithful Einstein–Boltzmann implementation and a Planck/BAO/SN/RSD/GW likelihood have not been run for RFG-R or RFG-RΞ.

The covariant KGB realization in Sec. 20 supplies a distinct, action-faithful photon–baryon–CDM–neutrino computation, an executed Planck CMB-plus-lensing fixed-point likelihood, an exact-background Pantheon+–DESI DR2 BAO–chronometer conditional record, a selected-point local screening gate, and a native RSD audit. It is proposed as a physical replacement candidate for the phenomenological ΛCDM description, not as a merely parametrized competitor. Its action identities, global scalar-stability scan, pre-recombination gate, local screening calculation, and numerical records are reproducible from the linked public package. The empirical boundary remains precise: the reported -2.5637653 conditional difference has fixed non-displayed cosmological and nuisance inputs, while RSD lacks a survey-complete likelihood. The exact joint posterior in Sec. 20.5 establishes the stated Planck–Pantheon+–DESI DR2 BAO–chronometer inference. The independent ΛCDM control posterior in Sec. 20.6 is clear data support for the analysis rather than a demand for numerical identity: it confirms that the same local likelihood and late-time data interface recover the expected standard benchmark when the model is reduced to the reference phenomenology. The production Bayesian-evidence calculation in Sec. 20.7 now closes the marginal-likelihood gate for this exact comparison: it gives $\Delta \log Z = -44.2797735$, favoring the local ΛCDM reference over the present KGB realization under the declared priors and probe set. Consequently, empirical replacement is not asserted. A broader matched model comparison, public release of the evidence artifacts, and official RSD-inclusive multi-probe inference remain necessary before the R-Universe KGB candidate could be promoted beyond this delimited status.

The intended gain over the existing cosmological model is therefore qualitative before it can become finally quantitative. ΛCDM is retained as the benchmark because it organizes observations with remarkable economy, but it does not by itself explain why the dark sector should be a constant vacuum term plus pressureless invisible matter, why the late-time acceleration scale is selected, or how the same description should emerge from a deeper relational substrate. The R-Universe construction tries to supply that missing layer: a primitive capacity field, a time-ordering map, a variational gravitational realization, an exactly luminal tensor sector, a controlled local-GR limit, explicit scalar-constraint audits, and a route to CMB and large-scale-structure likelihoods. This is a larger explanatory burden than the reference model carries. The price of carrying it is that every additional layer must survive its own mathematical and observational gate. That is why the manuscript gives both the positive construction and the negative restrictions: replacement means a more complete physical account that is also exposed to stronger tests, not a looser model with more freedom.

Accordingly, the present status is deliberately sharp. The paper provides more structure than the phenomenological reference model and gives the community enough definitions, code links, numerical products, and failure criteria to check that structure. It does not ask the reader to accept R-Universe because ΛCDM has been numerically defeated; in the completed KGB evidence comparison it has not. It asks the reader to evaluate whether the R-Universe framework is a credible path toward replacing ΛCDM by a deeper dynamical theory, and it states exactly which calculations must still be completed before that path could become a published empirical replacement claim.

This manuscript is offered as a rigorously delimited theoretical foundation, not as a completed theory of nature.

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- preferred-foliation dynamics,
- hyperbolic field equations,
- conservative and dissipative transport,
- gravitational-wave propagation,
- and late-time cosmological acceleration.

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Data Availability: No original observational or experimental datasets were generated during the present theoretical study. The RFG-R numerical illustrations, parameter values, generated tables, validation tests, canonical-scalar reduction, extended-EFT pure-gravity scalar audit, multi-fluid action reduction, and PPN likelihood factor are publicly available in the reproducibility package linked in Sec. 12. The RFG-R multi-fluid CMB/matter likelihood is a released computation specification, not an executed observational inference. The covariant KGB realization, its native H–EFTCAMB spectra, its executed fixed-point likelihood, and its limited conditional Planck profile with paired fixed-input ΛCDM reference are publicly linked in Sec. 20. The exact-chain posterior is released with its four chain files, numerical summary, deep point-cache audit, hash-bound production audit, and provenance-marked TeX source in the versioned archive linked in Sec. 20.5. The production evidence values reported in Sec. 20.7 are taken from the completed local dynamic nested-sampling JSON/checkpoint/log records for the KGB and ΛCDM runs. These evidence records are included in the public reproduction package for this manuscript: `https://github.com/MartinPetrasek123/R_Universe_Evidence_Audited`. The directly auditable evidence records are archived at `https://github.com/MartinPetrasek123/R_Universe_Evidence_Audited/tree/main/evidence/kgb` and `https://github.com/MartinPetrasek123/R_Universe_Evidence_Audited/tree/main/evidence/lcdm`. The machine-checkable SHA–256 manifest is provided at `https://github.com/MartinPetrasek123/R_Universe_Evidence_Audited/blob/main/EVIDENCE_MANIFEST.sha256`, with a human-readable audit summary at `https://github.com/MartinPetrasek123/R_Universe_Evidence_Audited/blob/main/EVIDENCE_MANIFEST.md`. The Planck likelihood distribution remains subject to its own license and is not redistributed.

For data-related inquiries, contact: `martin.petrasek@irsfp.org`

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